Linear Algebra and Analytic Geometry
Lecture Notes Summer 2026
University of Graz

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June 20, 2026

Chapter 1

Review

In this chapter, we will recall some important concepts from the first semester which we will need in this course. To be more precise, we will recall matrix representations of linear transformations between finite dimensional vector spaces.

1.1 Matrix representations

Given a field k. A linear transformation $f\colon V \to W$ between two finite dimensional k-vectorspaces is a function with the property

$$f(x+y) = f(x+y)$$
$$f(cx) = cf(x)$$

for any $x, y \in V$ and $c \in k$. Given a basis $\alpha = \{v_1, \ldots, v_n\}$ of V and $\beta = \{w_1, \ldots, w_m\}$ of W, we can find a (unique) matrix which *represents* this linear transformation. But what does this mean? What does it mean to *represent* a linear transformation?

1.1.1 Coordinate vectors

We start with an easier question. Let k be a field and V be a finite dimensional k-vectorspace with an *ordered* basis $\mathsf{B} = \{v_1, \ldots, v_n\}$. This means that

1. B *spans* V. That is, any vector $v \in V$ can be written as a linear combination of vectors in V. This means that for any $v \in V$, we can find scalars $a_1, \ldots, a_n$ such that

$$v = a_1 v_1 + \ldots + a_n v_n.$$

2. B *is linearly independent*. This means that if v can be written as a linear combination of vectors in B, then this representation is unique. That is, if we can write the same vector as

$$v = a_1 v_1 + \ldots + a_n v_n$$
$$v = b_1 v_1 + \ldots + b_n v_n$$

then we must have $a_1 = b_1, \ldots, a_n = b_n$.

Therefore, if we have a basis $\mathsf{B} = \{v_1, \ldots, v_n\}$ of the k-vectorspace V and if we can write $v \in V$ as $v = a_1v_1 + \ldots + a_nv_n$, then any information regarding this vector v is given by the scalars $a_1, \ldots, a_n$. We write this as follows.

Definition 1.1.1. We put

$$[v]_{\mathsf{B}} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

where $\mathsf{B} = \{v_1, \ldots, v_n\}$ is a basis of the k-vectorspace V and if we can write $v \in V$ as $v = a_1v_1 + \ldots + a_nv_n$. Note that the column vector $[v]_{\mathsf{B}}$ is an element of k^n. We call it the *coordinate vector of v with respect to* B.

Remark 1.1.2. Note that the coordinate vector of a vector depends on the basis. If one changes the basis, the coordinate vector may or may not change.

Example 1.1.3. Let $V = P_2(\mathbb{R})$ be the space of polynomials with coefficients in $\mathbb{R}$ with degree up to 2. Consider the ordered bases $\mathsf{B}_1 = \{1, x, x^2\}$ and $\mathsf{B}_2 = \{1 + x, 1 - x, x^2\}$ and the vector $p = 5 + x + 7x^2$. The question

Determine $[p]_{\mathsf{B}_1}$.

is the same as the question

Find $a_1, a_2, a_3 \in \mathbb{R}$ *such that* $p = a_1 \cdot 1 + a_2 \cdot x + a_3 \cdot x^2$.

Therefore, we have

$$[5 + x + 7x^2]_{\mathsf{B}_1} = \begin{bmatrix} 5 \\ 1 \\ 7 \end{bmatrix}.$$

Similarly, to determine $[p]_{\mathsf{B}_2}$, we should find $a, b, c \in \mathbb{R}$ such that

$$5 + x + 7x^2 = a(1 + x) + b(1 - x) + cx^2.$$

We can write the right hand side as $a + b + (a - b)x + cx^2$ and this gives us a system of linear equations

$$\begin{aligned} a + b &= 5 \\ a - b &= 1 \\ c &= 7 \end{aligned}$$

and by solving this system of linear equations, we find

$$[5 + x + 7x^2]_{\mathsf{B}_2} = \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}.$$

So, the same vector p has two different vector representations here.

Lemma 1.1.4. *The function $V \to k^n$ given by the rule $v \mapsto [v]_{\mathsf{B}}$ is a linear transformation. In fact, it is an isomorphism.*

Exercise 1.1.5. Prove this lemma.

Exercise 1.1.6. In class, we discussed that most of linear algebra is translating the language into correct form. Most of the time, by asking the question "What does it mean?" we can reduce our questions to solving a system of linear equations. Discuss the relationship between "B is a basis" and "$v \mapsto [v]_{\mathsf{B}}$ is an isomorphism" from this point of view.

1.1.2 Matrix of a linear transformation

Now, assume that we have a linear transformation $f\colon V \to W$ between two finite dimensional vectorspaces over a field k. Let $\mathsf{B} = \{v_1, \ldots, v_n\}$ be a basis for V and $\mathsf{C} = \{w_1, \ldots, w_m\}$ a basis for W. Then, by Lemma 1.1.4, we have a diagram

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "\sigma_{\mathsf{B}}"] & W \arrow[d, "\sigma_{\mathsf{C}}"] \\ k^n & k^m \end{tikzcd}$$

where σ_{B} and σ_{C} are isomorphisms. Since σ_{B} is an isomorphism, it has an inverse σ_{B}^{-1}. Hence, we get the diagram

$$\begin{tikzcd} V \arrow[r, "f"] & W \arrow[d, "\sigma_{\mathsf{C}}"] \\ k^n \arrow[u, "\sigma_{\mathsf{B}}^{-1}"] \arrow[r, dashed] & k^m \end{tikzcd} \tag{1.1.6.1}$$

where the dotted map is a linear map $k^n \to k^m$ given by the composition $\sigma_{\mathsf{C}} \circ f \circ \sigma_{\mathsf{B}}^{-1}$. This is again a linear map and we know from the first course in linear algebra that any such map is given by an $m \times n$ matrix with coefficients in k.

Definition 1.1.7. We call the matrix in the previous paragraph *the matrix representation of f with respect to* B *and* C. We denote it by $[f]_{\mathsf{B},\mathsf{C}}$.

This is a good definition but how do we actually write this matrix? Here, we will talk about the recipe, give an example and then discuss where this recipe comes from.

1. The first step in the recipe is to compute the images of the basis vectors $v_1, \ldots, v_n$ in the basis B. That is, compute $f(v_1), \ldots, f(v_n)$.
2. The vectors $f(v_1), \ldots, f(v_n)$ are elements of W. We know that that $\mathsf{C} = \{w_1, \ldots, w_m\}$ is a basis for C. So, write these elements as linear combinations of vectors in C. That is, find scalars a_{ij} such that

$$\begin{aligned} f(v_1) &= a_{11}w_1 + \ldots + a_{m1}w_m \\ f(v_2) &= a_{12}w_1 + \ldots + a_{m2}w_m \\ \vdots\ &\ \vdots \quad \ldots \quad \vdots \\ f(v_n) &= a_{1n}w_1 + \ldots + a_{mn}w_m \end{aligned}$$

3. Then, take the coordinate vectors $[f(v_1)]_{\mathsf{C}}, \dots, [f(v_n)]_{\mathsf{C}}$ and form the matrix $[f]_{\mathsf{B},\mathsf{C}}$ with these columns. That is, the matrix $[f]_{\mathsf{B},\mathsf{C}}$ is

$$[f]_{\mathsf{B},\mathsf{C}} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn}. \end{bmatrix}$$

Let us see this on an example.

Example 1.1.8. Let $V = P_1(\mathbb{R}) = \{a + bx \colon a, b \in \mathbb{R}\}$ be the space of polynomials of degree up to 1 with coefficients in $\mathbb{R}$. Consider the linear transformation

$$\begin{aligned} f \colon P_1(\mathbb{R}) &\to \mathbb{R}^3 \\ a + bx &\mapsto \begin{bmatrix} a+b \\ a-b \\ 2a+3b \end{bmatrix} \end{aligned}$$

We will write the matrix representation $[f]_{\mathsf{B},\mathsf{C}}$ where

$$\mathsf{B} = \{1 + x, 1 - x\} \text{ and } \mathsf{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

We will apply the three steps above. We compute

$$f(1+x) = \begin{bmatrix} 1+1 \\ 1-1 \\ 2+3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and find α, β, γ. We write this vector equation as a system of linear equations and solve that system of linear equations:

$$\begin{aligned} \alpha + \beta + \gamma &= 2 \\ \alpha + \beta &= 0 \\ \alpha &= 5 \end{aligned}$$

from which we get $\alpha = 5, \beta = -5, \gamma = 2$. These numbers will give us the first column. Next, we compute

$$f(1-x) = \begin{bmatrix} 1-1 \\ 1+1 \\ 2-3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} = r \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and get r, s, t. The corresponding system is

$$\begin{aligned} r + s + t &= 0 \\ r + s &= 2 \\ r &= -1 \end{aligned}$$

and the solution is $r = -1, s = 3, t = -2$. Hence, our matrix representation is

$$[f]_{\mathsf{B},\mathsf{C}} = \begin{bmatrix} 5 & -1 \\ -5 & 3 \\ 2 & -2 \end{bmatrix}.$$

Next, we will talk about the reason behind this recipe. And this goes back to the rules of matrix multiplication.

Exercise 1.1.9. Compute

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ \pi & 3 & \sqrt{2} & 41 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

This computation should take you less than 2 seconds. Because you should know by now that when we multiply by a matrix A with a standard basis vector e_i, then Ae_i gives you the ith column of A. So, in the above example, we have

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ \pi & 3 & \sqrt{2} & 41 \end{bmatrix} \begin{bmatrix} [f]_{\mathsf{B},\mathsf{C}}[v]_{\mathsf{B}} = [f(v)]_{\mathsf{C}} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ \pi \end{bmatrix}$$

And similarly, we have

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ \pi & 3 & \sqrt{2} & 41 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ \pi & 3 & \sqrt{2} & 41 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ \sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} [f]_{\mathsf{B},\mathsf{C}}[v]_{\mathsf{B}} = [f(v)]_{\mathsf{C}} 1 & 2 & 3 & 4 \\ \pi & 3 & \sqrt{2} & 41 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 41 \end{bmatrix}.$$

Theorem 1.1.10. *Let $f\colon V \to W$ be a linear transformation,* B *a basis for* V *and* C *a basis for* W*. Then, for any* $v \in V$*, we have*

$$[f]_{\mathsf{B},\mathsf{C}}[v]_{\mathsf{B}} = [f(v)]_{\mathsf{C}}.$$

Proof. By definition of our matrix representations (see the diagram 1.1.6.1), we have

$$\begin{tikzcd} V \arrow[r, "f"] \arrow[d, "\sigma_{\mathsf{B}}"'] & W \arrow[d, "\sigma_{\mathsf{C}}"] \\ k^n \arrow[r, dashed, "{[f]_{\mathsf{B},\mathsf{C}}}"] & k^m \end{tikzcd}.$$

This diagram tells us that we can start with $v \in V$, look at its coordinates (this is applying σ_{B}) and multiply this coordinate vector with the matrix representation $[f]_{\mathsf{B},\mathsf{C}}$ OR we can apply f to v and then look at the coordinates of $f(v)$. These two will give us the same thing. □

Exercise 1.1.11. Let U, V, W be three finite dimensional vectorspaces over a field k. Let B be a basis of U, C a basis of V and D a basis for W. And finally, let $f\colon U \to V$ and $g\colon V \to W$ be linear transformations. Prove that we have an equality

$$[g \circ f]_{\mathsf{B},\mathsf{D}} = [g]_{\mathsf{C},\mathsf{D}} \cdot [f]_{\mathsf{B},\mathsf{C}}.$$

Exercise 1.1.12. You have learned that $\mathbb{C}$ is a two dimensional vectorspace over $\mathbb{R}$ with basis $\mathsf{B} = \{1, i\}$.

1. Let $k\colon \mathbb{C} \to \mathbb{C}$ be the complex conjugation
$$k(x + yi) = x - yi.$$
Find $[k]_{\mathsf{B},\mathsf{B}}$.

2. Let $m_i\colon \mathbb{C} \to \mathbb{C}$ be the multiplication by i:
$$m_i(z) = iz.$$
Find $[m_i]_{\mathsf{B},\mathsf{B}}$.

Exercise 1.1.13. Let $P_1(\mathbb{R}) = \{a + bx\colon a, b \in \mathbb{R}\}$ be the vectorspace of polynomials with coefficients in $\mathbb{R}$ of degree ≤ 1. The linear map $f\colon P_1(\mathbb{R}) \to \mathbb{R}^3$ is given by

$$f(a + bx) = \begin{bmatrix} 2a + 3b \\ a + b \\ a - b \end{bmatrix}.$$

1. Compute $[f]_{\mathsf{B},\mathsf{C}}$ with bases
$$\mathsf{B} = \{1, 1 + x\} \quad , \quad \mathsf{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

2. Find $[v]_{\mathsf{B}}$ with $v = 2 + 3x$.

3. Find $[f(v)]_{\mathsf{C}}$ and $f(v)$ with $v = 2 + 3x$.

Exercise 1.1.14. Let V be the vectorspace of 2×2 matrices with coefficients in $\mathbb{R}$. Consider the linear map $f\colon V \to V$ given by the rule $f(A) = A + A^T$.

1. Determine the matrix $[f]_{\mathsf{B},\mathsf{B}}$ where
$$\mathsf{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

2. A matrix is called *symmetric* if $A^T = A$ holds. Compute $f(A)$ where A is a symmetric matrix.

3. A matrix is called *antisymmetric* if $A^T = -A$ holds. Compute $f(A)$ where A is an antisymmetric matrix.

4. Determine $[f]_{\mathsf{C},\mathsf{C}}$ with
$$\mathsf{C} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right\}.$$

1.1.3 Change of basis

Given a finite dimensional vectorspace V over a field k. We have seen that a basis B of V gives us an isomorphism $\sigma_{\mathsf{B}}\colon V \to k^n$ and this gives us the coordinate vector $[v]_{\mathsf{B}}$ of v with respect to the basis B. However, a basis of V is not uniquely determined. One can choose different bases and therefore, *may* get a different coordinate vector depending on the basis. So, if I have two bases B, C of the same finite dimensional vectorspace V, how are they related to each other? This is what we want to know in this subsection.

Our answer will rely on the diagram from the definition of a matrix representation. We have two bases B and C of the same n-dimensional vectorspace V. Hence, we have two maps

$$\begin{tikzcd} V \arrow[d, "\sigma_{\mathsf{B}}"'] & V \arrow[d, "\sigma_{\mathsf{C}}"'] \\ k^n & k^n \end{tikzcd}$$

We want to know the image of the same vector v under these linear maps σ_B and σ_{C}. This is saying the same thing as we want to know the coordinates of v with respect to B and C. Since, we are interested in the same vector on both left and right, we can represent this simply as

$$\begin{tikzcd} V \arrow[r, "\mathrm{id}"] \arrow[d, "\sigma_{\mathsf{B}}"'] & V \arrow[d, "\sigma_{\mathsf{C}}"'] \\ k^n & k^n \end{tikzcd}$$

where id is the identity transformation. So, the corresponding map

$$\begin{tikzcd} V \arrow[r, "\mathrm{id}"] \arrow[d, "\sigma_{\mathsf{B}}"'] & V \arrow[d, "\sigma_{\mathsf{C}}"'] \\ k^n \arrow[r, dashed] & k^n \end{tikzcd}$$

given by the dotted arrow is what tells us how the coordinate vector changes. And this is exactly the matrix representation of the identity map from V to itself with respect to the bases B and C.

Definition 1.1.15. The change of basis matrix from B to C is the matrix representation $[\mathrm{id}]_{\mathsf{B},\mathsf{C}}$ of the identity map $\mathrm{id}\colon V \to V$ with respect to the bases B, C.

Let us see an example.

Example 1.1.16. Let $V = P_2(\mathbb{R}) = \{a + bx + cx^2 \colon a, b, c \in \mathbb{R}\}$. Consider the bases

$$\mathsf{B} = \{1, 1 + x, 1 + x + x^2\} \text{ and } \mathsf{C} = \{1 + x + x^2, x + x^2, x^2\}.$$

In order to find the change of basis matrix from B to C. We will do the same steps we did to compute the matrix representation of some arbitrary function, but with the identity function. So, we will apply the identity function to the elements of the first basis and then we look at the image and we will write the image with respect to the second basis. But, the identity matrix doesn't do anything. So, the image is the same as the original basis vectors. We have

$$\begin{aligned} 1 &= a(1 + x + x^2) + b(x + x^2) + cx^2 \\ 1 + x &= d(1 + x + x^2) + e(x + x^2) + fx^2 \\ 1 + x + x^2 &= g(1 + x + x^2) + h(x + x^2) + ix^2. \end{aligned}$$

The first line gives us $1 = a + (a + b)x + (a + b + c)x^2$ and thus

$$\begin{aligned} a &= 1 \\ a + b &= 0 \\ a + b + c &= 0 \end{aligned}$$

which gives us $a = 1, b = -1, c = 0$. The second line gives us

$$1 + x = d + (d + e)x + (d + e + f)x^2$$

which tells us

$$\begin{aligned} d &= 1 \\ d + e &= 1 \\ d + e + f &= 0 \end{aligned}$$

and we have $d = 1, e = 0, f = -1$. Finally, it is easy to see that $g = 1, h = i = 0$. So, our matrix is

$$[\mathrm{id}]_{\mathsf{B},\mathsf{C}} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

Exercise 1.1.17. Show that the change of basis matrix is invertible. What is the inverse?

Exercise 1.1.18. Let K be a field, V a K-vectorspace and $\mathsf{B} = (v_1, v_2, v_3, v_4), \mathsf{C} = (v_2, v_3, v_4, v_1)$ two bases.

1. Determine the change of basis matrix from B to C.
2. Let A be the change of basis matrix you found in the previous part. Compute A^2, A^3, A^4 and A^{2025}.

Exercise 1.1.19. Let $V = M_{2\times 2}(\mathbb{R})$ be the space of 2×2 matrices with coefficients in $\mathbb{R}$ and

$$\mathsf{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \quad \text{and}$$
$$\mathsf{C} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right\}$$

two bases of V.

1. Compute the change of basis matrix from C to B.
2. Compute the change of basis matrix from B to C.

1.2 Similarity

We have seen how two representations of the same vector are related by the change of basis matrix. Now, we will see how two representations of the same linear map are related in this subsection.

More precisely, we will consider a finite dimensional vectorspace V with bases B, C and a linear function $f\colon V \to V$. We learned how to construct the matrix representations $[f]_{\mathsf{B},\mathsf{B}}$ and $[f]_{\mathsf{C},\mathsf{C}}$. We are asking how these two square matrices are related.

Here, it is useful to think of a linear map from V to itself as an abstract process or as a machine. Then, the matrix representation is how we explain or translate this process to a machine. The matrix $[f]_{\mathsf{A},\mathsf{D}}$ for two bases A, D of V takes, as input, something that is in terms of the vectors in A. We can think this as the A-language. Then, it gives an output in terms of D or in the D-language.

Remember that we are dealing with two matrix representations $[f]_{\mathsf{B},\mathsf{B}}$ and $[f]_{\mathsf{C},\mathsf{C}}$. One of them entirely works in the B-language: takes inputs in the B-language and gives outputs in the B-language. The other one does this in the C-language. So, if we have a vector $v \in V$ and its coordinate vector $[v]_{\mathsf{B}}$ (the vector v in the B-language) and if we want to give this vector as an input to the $[f]_{\mathsf{C},\mathsf{C}}$-machine, we first need to translate the coordinate vector from the B-language to the C-language. Then, it is possible for it to be taken as input into the $[f]_{\mathsf{C},\mathsf{C}}$-machine. The result this machine gives is again in the C-language. Translating back to the B-language gives us the following theorem.

Theorem 1.2.1. *Let V be a finite dimensional vectorspace and B, C be bases of V. Then,* *we have the change of basis formula*

$$[f]_{\mathsf{B},\mathsf{B}} = [\mathrm{id}]_{B,C}^{-1} \cdot [f]_{\mathsf{C},\mathsf{C}} \cdot [\mathrm{id}]_{B,C}.$$

Proof. We have already explained in the above paragraph how the translation should work. Then, it only remains to use Exercise 1.1.17. □

Theorem 1.2.1 tells us that if A and B are two square matrices which represent the same linear transformation, then there exists an invertible matrix S such that $A = S^{-1}BS$. We give this a name.

Definition 1.2.2. Let A and B be two $n \times n$ matrices with entries in a field K. We say that A and B are *similar* if there exists an invertible matrix S such that

$$A = S^{-1}BS.$$

Exercise 1.2.3. Show that Theorem 1.2.1 can be generalised as follows. Let $f\colon V \to W$ be a linear transformation between two finite dimensional vectorspaces V and W. Let B, B' be bases for V and C, C' be bases for W. Show that

$$[f]_{\mathsf{B},\mathsf{C}} = [\mathrm{id}]_{\mathsf{C}',\mathsf{C}} \cdot [f]_{\mathsf{B}',\mathsf{C}'} \cdot [f]_{\mathsf{B},\mathsf{B}'}.$$

Exercise 1.2.4. Let K be a field and $V = M_{n\times n}(K)$ be the vectorspace of $n \times n$ matrices with coefficients in K. Show that the similarity of matrices is an equivalence relation on V.

Exercise 1.2.5. Let

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \text{ und } B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Is A similar to B? If yes, which invertible satisfies $B = S^{-1}AS$? If no, explain why.

Exercise 1.2.6. Let K be a field and A, B two $n \times n$ matrices with coefficients in K. **True or False.**

1. If A is similar to B and A is invertible, then B is also invertible.
2. The space of ll matrices which are similar to A is a subspace.
3. If A is similar to the zero matrix, then A is the zero matrix.
4. If there is only one matrix similar to A, then A is the zero matrix.

Exercise 1.2.7. Let K be a field and A be an $n \times n$ matrix with coefficients in K. The *trace* of A is defined as the sum of the diagonal entries. We denote it by $\mathrm{tr}(A)$.

1. For two $n \times n$ matrices A and B, compute the diagonal entries of AB and BA. Show that there is an equality $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.

2. Show that similar matrices have the same trace.
3. Let

$$A = \begin{bmatrix} 2 & \pi & \sqrt{2} \\ 71 & 3 & 1 \\ 72 & 43 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & \sqrt{3} & \sqrt{23} \\ 7 & 3 & 8 \\ 21 & 45 & 1 \end{bmatrix}.$$

   Is A similar to B?
4. Let

$$C = \begin{bmatrix} 3 & \sqrt{3} & \sqrt{23} \\ 7 & 3 & 0 \\ 14 & 6 & 0 \end{bmatrix}$$

   Is A similar to C?

Exercise 1.2.8. Let $K = \mathbb{Z}/2\mathbb{Z}$ be the finite field with two elements $\{0, 1\}$ and V be the vectorspace of 2×2 matrices with coefficients in K. There are 16 matrices in V and the trace of a matrix is either 0 or 1.

1. Determine all the matrices A with $\mathrm{tr}(A) = 0$.
2. Determine all invertible matrices A with $\mathrm{tr}(A) = 0$.
3. Determine all the matrices A with $\mathrm{tr}(A) = 1$.
4. Determine all invertible matrices A with $\mathrm{tr}(A) = 1$.
5. Find all similarity classes of matrices in V.

Chapter 2

Inner Product and Norm

In this chapter, we are going to do some geometry using linear algebra.

Exercise 2.0.1. In order to do plane geometry, what notions do you need?

Before doing this, we are going to introduce the notion of an adjoint matrix.

2.1 Adjoint matrices

In order to define the adjoint matrix, we need to make sure that we know two important concepts: the nspose of a matrix and the complex conjugation.

Exercise 2.1.1. Before continuing, make sure that you know the transpose of a matrix and the complex conjugation.

Definition 2.1.2. Let A be an $m \times n$ matrix with coefficients in $\mathbb{C}$. Then, the *adjoint matrix* to A is defined as $A^* = (\overline{A})^T$.

Let us quickly talk about the definition. The notation $\overline{A}$ means that we should take the (complex) conjugate of all entries in A and after that we need to take the transpose.

Exercise 2.1.3. Let A be an $m \times n$ matrix with coefficients in $\mathbb{C}$.

1. Show that $\overline{(\overline{A})} = A$.
2. Show that $(A^T)^T = A$.
3. Show that $(\overline{A})^T = \overline{(A^T)}$.
4. Now using these three, show that $(A^*)^* = A$.

Exercise 2.1.4. Let A, B be an $m \times n$ matrix with coefficients in $\mathbb{C}$ and $c \in \mathbb{C}$.

1. Show that $(A + B)^* = A^* + B^*$.
2. Show that $(cA)^* = \overline{c}A^*$.

Exercise 2.1.5. Let A be an $m \times n$ matrix and B an $n \times k$ matrix with coefficients in $\mathbb{C}$. Show that $(AB)^* = B^* A^*$.

Exercise 2.1.6. Let A be an invertible matrix with coefficients in $\mathbb{C}$. Show that A^* is also invertible and $(A^{-1})^* = (A^*)^{-1}$.

Exercise 2.1.7.

1. Let V be the $\mathbb{R}$-vectorspace of 2×2 matrices with coefficients in $\mathbb{R}$. Is
$$U = \{A \in V : A^T = A\}$$
a subspace of V?

2. Let W be the $\mathbb{C}$-vectorspace of 2×2 matrices with coefficients in $\mathbb{C}$. Is
$$T = \{A \in V : A^* = A\}$$
a subspace of W?

2.2 Inner Product

In order to do plane geometry in $\mathbb{R}^2$, one thing we need is the concept of angles. An inner product (or scalar product) is what will provide it for us.

Definition 2.2.1. Let $K = \mathbb{R}$ or $K = \mathbb{C}$ and V be a vectorspace over K. An *inner product* or *scalar product* on V is a map
$$\langle \cdot, \cdot \rangle : V \times V \to K$$
such that for any $u, v, w \in V$ and for all $c \in K$, we have

1. $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$.
2. $\langle u, cv \rangle = c \langle u, v \rangle$.
3. $\langle u, v \rangle = \overline{\langle v, u \rangle}$.
4. $\langle u, u \rangle \geq 0$.

4,5. $\langle u, u \rangle = 0 \iff u = 0$.

Definition 2.2.2. A vectorspace over $\mathbb{R}$ with an inner product is called a *Euclidean vectorspace*. A vectorspace over $\mathbb{C}$ with an inner product is called a *unitary vectorspace*. In this case, we write $(V, \langle \cdot, \cdot \rangle)$.

Remark 2.2.3. In a vectorspace over $\mathbb{R}$ or over $\mathbb{C}$, one can define different inner products. In fact, there are infinitely many ways to define an inner product.

Exercise 2.2.4. Show that in a Euclidean or a unitary vectorspace V, one has
$$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$
for any $u, v, w \in V$.

Exercise 2.2.5. Show that in a unitary vectorspace V, one has

$$\langle cu, v\rangle = \overline{c}\langle u, v\rangle$$

for any $u, v \in V$ and $c \in \mathbb{C}$.

These two exercises show that while the inner product is linear in the second coordinate (thanks to the first two axioms of an inner product), it is not linear in the first coordinate if the field is $K = \mathbb{C}$.

Exercise 2.2.6. Show that in a Euklidean vectorspace V, one has

$$\langle cu, v\rangle = \langle u, v\rangle$$

for any $u, v \in V$ and $c \in \mathbb{R}$.

This shows that the inner product is *bilinear* on a Euclidean vectorspace (when the field is $K = \mathbb{R}$). Bilinear means linear in both coordinates.

Exercise 2.2.7. Show that in a Euclidean or a unitary vectorspace V, one has

$$\langle u, 0\rangle = \langle 0, u\rangle = 0$$

for any $u \in V$.

Remark 2.2.8. Note that in this exercise, we denote two different things by the symbol 0. It is clear from the context that the 0 inside the brackets is the zerovector and the 0 outside is the number 0. In this course, we are not going to use separate notation if the context is clear.

Example 2.2.9 (Standard inner product on $\mathbb{R}^n$)**.** For

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \text{ and } y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n,$$

the rule

$$\langle x, y\rangle = x_1y_1 + \ldots + x_ny_n$$

defines an inner product which we call the *standard inner product* on $\mathbb{R}^n$.

Exercise 2.2.10. Show that the standard inner product on $\mathbb{R}^n$ actually satisfies the axioms of an inner product.

Remark 2.2.11. We can not define the standard inner product on $\mathbb{C}^n$ using the same rule. For instance, for $n = 2$ and

$$x = \begin{bmatrix} i \\ i \end{bmatrix}$$

this rule would give us $\langle x, x\rangle = i^2 + i^2 = -2$. Why is this an issue?

Example 2.2.12 (Standard inner product on $\mathbb{C}^n$)**.** In order to fix the issue in the Remark, we define the standard inner product on $\mathbb{C}^n$ as

$$\langle x, y\rangle = \overline{x_1}y_1 + \ldots + \overline{x_n}y_n.$$

Exercise 2.2.13. Show that the standard inner product on $\mathbb{C}^n$ actually satisfies the axioms of an inner product.

Example 2.2.14. Let $a, b \in \mathbb{R}$ and $V = \{f : [a, b] \to \mathbb{R} | f \text{ is continuous}\}$ be the vectorspace of continuous functions on the interval $[a, b]$. Then, the rule

$$\langle f, g\rangle = \int_a^b f(x)g(x)dx$$

defines an inner product on V. The linearity in the second coordinate follows from the properties of the integral operation. The symmetry $\langle f, g\rangle = \langle g, f\rangle$ follows from the fact that the multiplication in $\mathbb{R}$ is commutative (why? Explain this as an exercise). Moreover, since for any $x \in [a, b]$ and any function f on $[a, b]$, we have $[f(x)]^2 \geq 0$. This shows that $\langle f, f\rangle \geq 0$ for any f. Finally, since f is continuous, we know that so is f^2. If there exists an $x \in [a, b]$ with $f^2(x) > 0$, then by continuity of f^2, there will be a positive area under the graph of f^2 near x. This is all we needed to show.

Example 2.2.15. Choosing a basis of a finite dimensional vectorspace allows us to define an inner product, too. Let V be an n-dimensional vectorspace over $\mathbb{R}$ with basis B. Then, the rule

$$\langle x, y\rangle = \langle [x]_{\mathsf{B}}, [y]_{\mathsf{B}}\rangle$$

defines an inner product on V where the inner product on the right hand side is the standard inner product on the right hand side. For instance, let $V = P_1(\mathbb{R})$ be the space of polynomials with real coefficients with degrees up to 1 and let $\mathsf{B} = \{1, x\}$ be a basis for V. Then, we have

$$\langle a + bx, c + dx\rangle = \langle \begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} c \\ d \end{bmatrix}\rangle = ac + bd$$

is an inner product on V. As an exercise, prove this statement.

The following exercise is a more general version of the previous example.

Exercise 2.2.16. Let V be a real vectorspace and $(W, \langle -, -\rangle_W)$ be a Euclidean space and let $f : V \to W$ be an injective linear map. Define

$$\langle x, y\rangle_V := \langle f(x), f(y)\rangle_W$$

for all $x, y \in V$. Show that this defines an inner product on V.

Example 2.2.17. Consider the space of $n \times n$ matrices over $\mathbb{C}$. Then, the rule

$$\langle A, B\rangle = \mathrm{tr}(A^*B)$$

defines an inner product on R. One way to show this is to compute that

$$\begin{bmatrix} a & c \\ b & d \end{bmatrix}^* \begin{bmatrix} x & z \\ y & t \end{bmatrix} = \begin{bmatrix} \bar{a} & \bar{b} \\ \bar{c} & \bar{d} \end{bmatrix} \begin{bmatrix} x & z \\ y & t \end{bmatrix} = \begin{bmatrix} \bar{a}x + \bar{b}y & \cdots \\ \cdots & \bar{c}z + \bar{d}t \end{bmatrix}$$

and to observe that

$$\langle \begin{bmatrix} a & c \\ b & d \end{bmatrix}, \begin{bmatrix} x & z \\ y & t \end{bmatrix} \rangle = \bar{a}x + \bar{b}y + \bar{c}z + \bar{d}t.$$

So, this is a version of Example 2.2.15. Explain this last sentence.

2.3 Norm

Next, we are going to talk about norm of a vector. This will play the role of a length or magnitude of a vector. So, there are certain things that we want from it such as it has to be a real number and it has to be non-negative. Also, we want to do geometry and one of the simplest things we can require is the triangle inequality.

Definition 2.3.1. Let V be a vectorspace over $K = \mathbb{R}$ or over $K = \mathbb{C}$. A *norm* on V is a mapping $\|\cdot\|: V \to \mathbb{R}$ such that for any $u, v \in V$ and $c \in K$ we have

1. $\|u\| \geq 0$ and $\|u\| = 0$ if and only if $u = 0$.
2. $\|cu\| = |c| \cdot \|u\|$.
3. $\|u + v\| \leq \|u\| + \|v\|$ (Triangle inequality).

A vectorspace with a norm is called a normed vectorspace.

Example 2.3.2. Let $V = \mathbb{R}^n$ and p be a positive integer. Then,

$$\text{for } v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \text{ we put } \|v\|_p := \left(x_1^p + \ldots + x_n^p\right)^{1/p}.$$

We claim that this is a norm. The first two properties of the norm is relatively easier to show. The third property is challenging but you are assumed to know this from earlier courses (Analysis).

Exercise 2.3.3. Let $V = \mathbb{R}^n$ and

$$\text{for } v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \text{ define } \|v\|_\infty := \max_{1 \leq i \leq n} |x_i|.$$

Show that this defines a norm.

Exercise 2.3.4. For $p = 1, 2, 3, \infty$. Draw the *unit circle*

$$\left\{x \in \mathbb{R}^2 : \|x\|_p = 1\right\}.$$

Exercise 2.3.5. Let V be the space of $n \times n$ matrices with entries in $\mathbb{C}$. For $A \in V$, define

$$\|A\| := \max_{\|v\|=1} \|Av\|_2.$$

Show that this defines a norm.

Norm allows us to define distances in an abstract vectorspace.

Definition 2.3.6. Let V be a normed vectorspace and $x, y \in V$. The *distance* between x and y is defined to be $\|x - y\|$.

We now have more geometric theorems.

Lemma 2.3.7. *Let V be a normed vectorspace. Then, for any $x, y \in V$, we have*

$$|\|x\| - \|y\|| \leq \|x - y\| \leq \|x\| + \|y\|.$$

Proof. We start with the first inequality. The main key is the simple fact that $y - y = 0$. This gives us $x - y + y = x$. Therefore, we have

$$\|x\| = \|x - y + y\| \leq \|x - y\| + \|y\|$$

by triangle inequality and thus

$$\|x\| - \|y\| \leq \|x - y\|.$$

On the other hand, simply by replacing the roles of x nd y, we have

$$\|y\| - \|x\| \leq \|y - x\| = \|(-1)(x - y)\| = |(-1)|\|x - y\| = \|x - y\|.$$

Combining these two we get

$$|\|x\| - \|y\|| \leq \|x - y\|$$

as required. For the second inequality, we simply have

$$\|x - y\| = \|x + (-1)y\| \leq \|x\| + \|(-1)y\| = \|x\| + |-1|\|y\| = \|x\| + \|y\|$$

□

Exercise 2.3.8. Let V be a normed vectorspace. Show that for any $v_1, \ldots, v_n \in V$ and $c_1, \ldots, c_n \in K$, we have

$$\|c_1 v_1 + \ldots + c_n v_n\| \leq |c_1|\|v_1\| + \ldots + |c_n|\|v_n\|.$$

Theorem 2.3.9 (Cauchy-Schwarz Inequality)**.** *Let V be a Euclidean or unitary vectorspace. Then, for any $u, v \in V$, we have*

$$|\langle x, y\rangle| \leq \sqrt{\langle x, x\rangle}\sqrt{\langle y, y\rangle}.$$

The equality holds if and only if x, y are linearly dependent.

Proof. Before starting the proof, let us note that the right hand side makes sense because the inner product of a vector with itself is a real number and it is not negative by definition of an inner product.

We will consider two cases. Firstly, assume that x, y are linearly dependent. Then, without loss of generality, there is a $c \in K$ (here $K = \mathbb{R}$ or $\mathbb{C}$) such that $y = cx$. Then, on the one hand we have

$$|\langle x, y\rangle| = |\langle x, cx\rangle| = |c\langle x, x\rangle| = |c|\langle x, x\rangle.$$

and on the other hand,

$$\begin{aligned}\sqrt{\langle x, x\rangle}\sqrt{\langle y, y\rangle} = \sqrt{\langle x, x\rangle}\sqrt{\langle cx, cx\rangle} &= \sqrt{\langle x, x\rangle}\sqrt{\bar{c}\cdot c\cdot\langle x, x\rangle}\\ &= \sqrt{\langle x, x\rangle}\sqrt{|c|^2\langle x, x\rangle}\\ &= |c|\langle x, x\rangle.\end{aligned}$$

This, we have the desired equality.

Now, assume that x and y are linearly independent. Then, for any $\lambda \in K$, we have $\lambda x - y \neq 0$. Hence, we have

$$\langle \lambda x - y, \lambda x - y\rangle > 0.$$

Note that for all $\lambda \in K$, we have

$$\begin{aligned}\langle \lambda x - y, \lambda x - y\rangle &= \langle \lambda x, \lambda x\rangle + \langle \lambda x, -y\rangle + \langle -y, \lambda x\rangle + \langle -y, -y\rangle\\ &= \bar{\lambda}\cdot\lambda\langle x, x\rangle - \bar{\lambda}\langle x, y\rangle - \lambda\langle y, x\rangle + (-1)^2\langle y, y\rangle\end{aligned}$$

Since this holds for all λ, it must hold for a specific λ that we will choose. In order to cancel things out, we will put

$$\lambda := \frac{\langle x, y\rangle}{\langle x, x\rangle}.$$

Note that with this and using properties of inner product, we get

$$\bar{\lambda} = \overline{\frac{\langle x, y\rangle}{\langle x, x\rangle}} = \frac{\langle y, x\rangle}{\langle x, x\rangle}.$$

Therefore, we have

$$\begin{aligned}\langle \lambda x - y, \lambda x - y\rangle = &= \bar{\lambda}\cdot\lambda\langle x, x\rangle - \bar{\lambda}\langle x, y\rangle - \lambda\langle y, x\rangle + (-1)^2\langle y, y\rangle\\ &= \frac{\langle y, x\rangle}{\langle x, x\rangle}\cdot\frac{\langle x, y\rangle}{\langle x, x\rangle}\cdot\langle x, x\rangle - \frac{\langle y, x\rangle}{\langle x, x\rangle}\langle x, y\rangle - \frac{\langle x, y\rangle}{\langle x, x\rangle}\langle y, x\rangle + \langle y, y\rangle\\ &= \frac{\langle y, x\rangle\langle x, y\rangle}{\langle x, x\rangle} - \frac{\langle y, x\rangle\langle x, y\rangle}{\langle x, x\rangle} - \frac{\langle x, y\rangle\langle y, x\rangle}{\langle x, x\rangle} + \langle y, y\rangle.\end{aligned}$$

Hence, we get

$$\begin{aligned}
\langle \lambda x - y, \lambda x - y\rangle > 0 &\iff -\frac{\langle x,y\rangle\langle y,x\rangle}{\langle x,x\rangle} + \langle y,y\rangle > 0\\
&\iff \frac{\langle x,y\rangle\langle y,x\rangle}{\langle x,x\rangle} < \langle y,y\rangle\\
&\iff \langle x,y\rangle\langle y,x\rangle < \langle x,x\rangle \cdot \langle y,y\rangle\\
&\iff \langle x,y\rangle\overline{\langle x,y\rangle} < \langle x,x\rangle \cdot \langle y,y\rangle\\
&\iff |\langle x,y\rangle|^2 < \langle x,x\rangle \cdot \langle y,y\rangle
\end{aligned}$$

Therefore, we have

$$|\langle x,y\rangle| < \sqrt{\langle x,x\rangle}\sqrt{\langle y,y\rangle}.$$

□

The Cauchy-Schwarz inequality gives us a lot of cool things. First of all, it is very helpful in Analysis to compute or bound some integrals. But secondly, it allows us to define things in this course.

Definition 2.3.10. Let V be a Euclidean or unitary space with inner product $\langle -,-\rangle$. Then,

$$\|x\| := \sqrt{\langle x,x\rangle}$$

defines a norm on V. We call this norm the *induced norm*.

Exercise 2.3.11. Show that the axioms of a norm is satisfied for

$$\|x\| := \sqrt{\langle x,x\rangle}.$$

Hint: You need to use Cauchy-Schwarz inequality at some point.

Remark 2.3.12. Note that with this definition, the Cauchy-Schwarz inequality reads

$$|\langle x,y\rangle| \leq \|x\| \cdot \|y\|.$$

We can make one more definition thanks to the Cauchy-Schwarz inequality.

Definition 2.3.13. Let V be a Euclidean vectorspace and $\|\cdot\|$ be the induced norm. Then the *angle* between two nonzero vectors $x, y \in V$ is the unique number $\varphi \in [0, 2\pi)$ with

$$\cos\varphi = \frac{\langle x,y\rangle}{\|x\| \cdot \|y\|}.$$

Now, this definition will allow us to talk about a whole new section. But before the next section, some exercises.

Exercise 2.3.14. Let V be the space of 2×2 matrices with coefficients in $\mathbb{C}$. We have learned that

$$\langle A, B \rangle = \operatorname{tr}(A^* B)$$

defines an inner product on V. Consider the induced norm.

1. For which values $a, b, c, d \in \mathbb{Z}$, does the matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

   satisfy $\|A\| = 1$?

2. True or False: For any $A \in V$, we have $\|A\| = \|A^*\|$.

2.4 Orthogonality

We have now defined everything that we need to prove a geometric theorem: The Pythagoras Theorem. We are going to do this in an abstract vectorspace with an inner product. The key hypothesis in Pythagoras's Theorem is a triangle with a right angle. Note that for $\varphi = \pi/2$, we have $\cos\varphi = 0$. Hence, by looking at Definition 2.3.13, we can say that two nonzero vectors in a Euclidean vectorspace have a right angle between them if and only if $\langle x, y \rangle = 0$. This will be our definition of perpendicularity, or in a fancier word: orthogonality.

Definition 2.4.1. Let V be a Euclidean or unitary vectorspace. We say that two vectors x and y in V are *orthogonal* to each other if $\langle x, y \rangle = 0$.

With this definition, let us prove our theorem.

Theorem 2.4.2 (The Pythagoras Theorem)**.** *Let V be a Euclidean space with the induced norm. If x and y are two orthogonal vectors in V, then we have*

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2.$$

Proof. The proof follows immediately if we write down the definitions and use bilinearity (since we are over $\mathbb{R}$). We now that by the definition of the induced norm, we have

$$\|x - y\|^2 = \langle x - y, x - y \rangle = \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle = \|x\|^2 + \|y\|^2$$

as required. Note that $\langle x, y \rangle = 0 = \langle y, x \rangle$ by our orthogonality assumption. □

Exercise 2.4.3. Let V be a Euclidean space with the induced norm. Prove the cosinus rule:

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\| \cos\phi$$

where ϕ is the angle between x and y.

Definition 2.4.4. Let V be a unitary or Euclidean space and $U \subseteq V$ be a subset. We say that $x \in V$ is orthogonal to U if x is orthogonal to every vector in U. The set of all vectors which are orthogonal to U is called the orthogonal complement of U and we denote it by $U^\perp$. That is, we have

$$U^\perp = \{x \in V \colon \langle x, u\rangle = 0 \text{ for all } u \in U\}.$$

Example 2.4.5. Consider $\mathbb{R}^3$ with the standard inner product

$$\left\langle \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right\rangle = ax + by + cz$$

and put

$$U = \left\{ \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix} : t \in \mathbb{R} \right\}.$$

We will determine $U^\perp$. For

$$v = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

we have

$$\begin{aligned} v \in U^\perp &\iff \langle v, u\rangle = 0 \text{ for all } u \in U \\ &\iff \left\langle \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix} \right\rangle = 0 \text{ for all } t \in \mathbb{R} \\ &\iff a \cdot t + b \cdot 0 + c \cdot 0 = 0 \text{ for all } t \in \mathbb{R} \\ &\iff at = 0 \text{ for all } t \in \mathbb{R} \\ &\iff a = 0. \end{aligned}$$

Hence, we conclude

$$U^\perp = \left\{ \begin{bmatrix} 0 \\ b \\ c \end{bmatrix} : b, c \in \mathbb{R} \right\}$$

We will now prove some properties of orthogonal complements or give as exercises.

Proposition 2.4.6. *Let V be Euclidean or unitary space. Then, for any subset $U \subset V$, the orthogonal complement $U^\perp$ is a subspace.*

Proof. By Exercise 2.2.7, we know that $0 \in U^\perp$. We need to show that $U^\perp$ is closed under addition and scalar multiplication. To show that it is closed under addition, let $x, y \in U^\perp$. Then, for any $u \in U$, we have

$$\langle x+y, u\rangle = \langle x, u\rangle + \langle y, u\rangle = 0 + 0 = 0$$

which shows $x + y \in U^\perp$. Similarly, for any $c \in K$ (with $K = \mathbb{R}$ or $\mathbb{C}$) and $x \in U^\perp$, we have

$$\langle cx, u\rangle = \bar{c}\langle x, u\rangle = \bar{c} \cdot 0 = 0 \text{ for all } u \in U.$$

□

Exercise 2.4.7. Let V be a Euclidean or unitary space and U be a subspace. Prove the following statements:

1. $U \cap U^\perp = \{0\}$.
2. $U \subseteq (U^\perp)^\perp$.
3. If $U \subseteq W \subseteq V$, then $W^\perp \subseteq U^\perp$.

Next, we will see how orthogality is helpful for us. We will start by illustrating this in a two dimensional Euklidean space. Assume that we have a two dimensional Euklidean space V with basis $\mathsf{B} = \{x, y\}$. Then, any vector $v \in V$ can be written uniquely as $v = ax + by$ for some $a, b \in \mathbb{R}$. Now, let $v = ax + by$ and $w = cx + dy$ be two vectors in V. Then, we have

$$\begin{aligned} \langle v, w\rangle &= \langle ax + by, cx + dy\rangle \\ &= ac\langle x, x\rangle + (ad + bc)\langle x, y\rangle + bd\langle y, y\rangle. \end{aligned}$$

On the other hand, we have

$$[v]_{\mathsf{B}} = \begin{bmatrix} a \\ b \end{bmatrix} \text{ and } [w]_{\mathsf{B}} = \begin{bmatrix} c \\ d \end{bmatrix}$$

and the standard inner product of these two vectors in $\mathbb{R}^2$ gives

$$\left\langle \begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} c \\ d \end{bmatrix} \right\rangle = ac + bd.$$

We observed

$$\begin{aligned} \langle v, w\rangle &= ac\langle x, x\rangle + (ad + bc)\langle x, y\rangle + bd\langle y, y\rangle \\ \langle [v]_{\mathsf{B}}, [w]_{\mathsf{B}}\rangle &= ac + bd. \end{aligned}$$

Hence, IF $\langle x, y\rangle = 0$ and IF $\langle x, x\rangle = \langle y, y\rangle = 1$, then the inner product of these two vectors would be the same as the inner product of their vector representations with respect to B. Of course, this does not work with every basis. But sometimes it happens and we give special names to these situations.

Definition 2.4.8. Let V be a Euklidean or a unitary vector space with norm. Let M be a nonempty subset of V and assume that M does not contain the zero vector.

1. We say that M is *an orthogonal set* if $\langle x, y\rangle = 0$ for any $x, y \in M$ with $x \neq y$.
2. We say that M is *an orthonormal set* if it is an orthogonal set with $\|x\| = 1$ for every $x \in M$.
3. We say that M is *an orthonormal basis* if it is an orthonormal set which is a basis of V.

Example 2.4.9. Let $V = \mathbb{R}^n$ with the standard inner product and let $\mathsf{B} = \{e_1, \ldots, e_n\}$ be the standard basis of V. Then, B is an orthonormal basis of V.

1. Each of the following sets form an orthonormal basis for $\mathbb{R}^4$:

   (a)
$$\left\{\begin{bmatrix}-1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\-1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\-1\end{bmatrix}\right\}$$

   (b)
$$\left\{\begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\-1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\-1\end{bmatrix}\right\}$$

2. For each $a, b \in \mathbb{R}$ with $a^2 + b^2 = 1$, that is for any point (a, b) on the unit circle in the plane, the set
$$\left\{\begin{bmatrix}a\\b\end{bmatrix}, \begin{bmatrix}-b\\a\end{bmatrix}\right\}$$
   forms an orthonormal basis of $\mathbb{R}^2$.

Example 2.4.10. Let $V = \{f : [0, 2\pi] \to \mathbb{R} | f \text{ is continuous}\}$ be the space of continuous functions on $[0, 2\pi]$ with the inner product
$$\langle f, g\rangle = \int_0^{2\pi} f(x)g(x)dx.$$

For any $n \geq 1$, consider the function f_n defined by the rule $f_n(t) := \dfrac{\sin(nt)}{\sqrt{\pi}}$. Then, the set
$$M = \{f_n : n \in N\}$$
is an orthonormal set in V. Note that it is not a basis.

Theorem 2.4.11. *Let V be a Euklidean or unitary vectorspace. Then, any orthogonal set in V is linearly independent.*

Proof. We will write the proof for the case of an orthogonal set with three elements to illustrate the idea of the proof and leave the general case as an exercise. Let $\{v_1, v_2, v_3\}$ be an orthogonal set in V. We want to show that the only solution to the equation

$$a_1v_1 + a_2v_2 + a_3v_3 = 0$$

is $a_1 = a_2 = a_3 = 0$. We have $\langle v_1, 0\rangle = 0$. This gives us

$$\langle v_1, a_1v_1 + a_2v_2 + a_3v_3\rangle = 0.$$

By using bilinearity, we get

$$0 = \langle v_1, a_1v_1 + a_2v_2 + a_3v_3\rangle = a_1\langle v_1, v_1\rangle + a_2\langle v_1, v_2\rangle + a_3\langle v_1, v_3\rangle.$$

Since we assumed that our set is an orthogonal set, this simplifies to

$$a_1\langle v_1, v_1\rangle = 0.$$

However, by our definition, v_1 can not be zero. This tells us by definition of an inner product that we must have $\langle v_1, v_1\rangle \neq 0$. Therefore, the equation

$$a_1\langle v_1, v_1\rangle = 0.$$

gives us $a_1 = 0$. Hence,

$$a_1v_1 + a_2v_2 + a_3v_3 = 0$$

implies $a_1 = 0$ and consequently,

$$a_2v_2 + a_3v_3 = 0.$$

By applying the same method as above, we get that $a_2 = 0$ which leaves us with $a_3v_3 = 0$. And this gives us $a_3 = 0$ as v_3 can not be zero. This finishes the proof for the case of an orthogonal set with three elements. $\square$

Exercise 2.4.12. Give a proof of the theorem for an arbitrary finite orthogonal set.

Now that we have a definition, we will record our discussion before the definition as a theorem.

Theorem 2.4.13. *Let V be a Euklidean or unitary space and $\mathsf{B} = \{v_1, \ldots, v_n\}$ an orthonormal basis for V. Let $u, v \in V$ with*

$$[u]_{\mathsf{B}} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad \text{and } [v]_{\mathsf{B}} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Then, we have

1. $\langle u, v\rangle = a_1b_1 + a_2b_2 + \cdots + a_nb_n$.
2. $\langle v_i, v\rangle = b_i$.

Exercise 2.4.14. By revisiting the discussion before Definition 2.4.8, write a proof of this theorem.

Exercise 2.4.15. Why is an orthonormal basis good? With your own words, write a one paragraph explanation.

Our next goal is to create orthonormal bases from a given basis. This is known as the *Gram-Schmidt process.*

Theorem 2.4.16. *Let V be an n-dimensional Euklidean or unitary space and $\{a_1, \ldots, a_n\}$ be a basis of V. Then, there exists an orthonormal basis $\{b_1, \ldots, b_n\}$ of V which can be computed as below:*

$$
\begin{aligned}
b_1 &:= \frac{1}{\|a_1\|}a_1,\\
\tilde{b_i} &:= a_i - \sum_{j=1}^{i-1}\langle b_j, a_i\rangle b_j,\\
b_i &:= \frac{1}{\|\tilde{b_i}\|}b_i.
\end{aligned}
$$

Remark 2.4.17. If one only wants an orthogonal basis, then it is enough to perform the middle step. However, if you try to do this, you will see that the computations eventually force you to do the other steps as well in secret.

Proof. We will prove this by induction by showing that $\{b_1, \ldots, b_i\}$ is an orthonormal basis for the linear span of $\{a_1, \ldots, a_i\}$ for any $i = 1, \ldots, n$. When $i = 1$, we see that b_1 is only a scalar multiple of a_i. Therefore, b_1 and a_1 span the same subspace. Assume that $\{b_1, \ldots, b_i\}$ is an orthonormal basis for the linear span of $\{a_1, \ldots, a_i\}$ for some i. We want to show that $\{b_1, \ldots, b_{i+1}\}$ is an orthonormal basis for the linear span of $\{a_1, \ldots, a_{i+1}\}$. We know that a_{i+1} is not in the span of $\{a_1, \ldots, a_i\}$ as these $i + 1$ vectors are linearly independent. Therefore, a_{i+1} is not in the span of $\{b_1, \ldots, b_i\}$. In other words, for all scalars $\lambda_1, \ldots, \lambda_i$, we have

$$a_{i+1} \neq \lambda_1 b_1 + \ldots + \lambda_i b_i.$$

In particular, we can take λ_j as the number $\langle b_j, a_{i+1}\rangle$. Hence, we know that

$$a_{i+1} \neq \langle b_1, a_{i+1}\rangle b_1 + \ldots + \langle b_i, a_{i+1}\rangle b_i.$$

In other words,

$$a_{i+1} - (\langle b_1, a_{i+1}\rangle b_1 + \ldots + \langle b_i, a_{i+1}\rangle b_i) \neq 0.$$

Now, we put

$$\tilde{b}_{i+1} := a_{i+1} - (\langle b_1, a_{i+1}\rangle b_1 + \ldots + \langle b_i, a_{i+1}\rangle b_i)$$

to get $\|\tilde{b}_{i+1}\| \neq 0$ which means that b_{i+1} is well-defined. By our definition, $\tilde{b}_{i+1}$ is in the span of $\{b_1, \ldots, b_i, a_{i+1}\}$. And by our assumption, this implies that $\tilde{b}_{i+1}$ is in the span of $\{a_1, \ldots, a_i, a_{i+1}\}$. Therefore, b_{i+1} is in the span of $\{a_1, \ldots, a_i, a_{i+1}\}$. Combining all this with our assumption, we see that the $i+1$ vectors $b_1, \ldots, b_{i+1}$ are all in the span of $\{a_1, \ldots, a_i, a_{i+1}\}$. Therefore, to conclude our theorem, it is enough to show that $b_1, \ldots, b_{i+1}$ are linearly independent vectors. Because it would mean that they span an $i+1$-dimensional space inside the $i+1$-dimensional subspace spanned by $\{a_1, \ldots, a_{i+1}\}$.

To show that $b_1, \ldots, b_{i+1}$ are linearly independent vectors, we will show that they form an orthogonal set. This would show linear independence by Theorem 2.4.11. And by our induction hypothesis, we know that $\{b_1, \ldots, b_n\}$ is an orthonormal set. So, it is enough to show that

$$\langle b_j, b_{i+1}\rangle = 0$$

for all $j \leq i$. And to do this, it is enough to show that

$$\langle b_j, \tilde{b}_{i+1}\rangle = 0$$

by linearity in the second componant. So, let's show this!

$$\begin{aligned}\langle b_j, \tilde{b}_{i+1}\rangle &= \langle b_j, a_{i+1} - (\langle b_1, a_{i+1}\rangle b_1 + \ldots + \langle b_i, a_{i+1}\rangle b_i)\rangle \\ &= \langle b_j, a_{i+1}\rangle - \langle b_1, a_{i+1}\rangle\langle b_j, b_1\rangle - \langle b_2, a_{i+1}\rangle\langle b_j, b_2\rangle - \cdots - \langle b_i, a_{i+1}\rangle\langle b_j, b_i\rangle.\end{aligned}$$

Note that in this sum we have terms with $\langle b_j, b_k\rangle$. However our assumption tells us that $\langle b_j, b_k\rangle = 0$ if $j \neq k$ and $\langle b_j, b_j\rangle = 1$. Hence, we end up with

$$\begin{aligned}\langle b_j, \tilde{b}_{i+1}\rangle &= \langle b_j, a_{i+1} - (\langle b_1, a_{i+1}\rangle b_1 + \ldots + \langle b_i, a_{i+1}\rangle b_i)\rangle \\ &= \langle b_j, a_{i+1}\rangle - \langle b_1, a_{i+1}\rangle\langle b_j, b_1\rangle - \langle b_2, a_{i+1}\rangle\langle b_j, b_2\rangle - \cdots - \langle b_i, a_{i+1}\rangle\langle b_j, b_i\rangle \\ &= \langle b_j, a_{i+1}\rangle - \langle b_j, a_{i+1}\rangle\langle b_j, b_j\rangle \\ &= \langle b_j, a_{i+1}\rangle - \langle b_j, a_{i+1}\rangle = 0.\end{aligned}$$

This completes the proof. □

Exercise 2.4.18. Write the procedure in the proof in your own words one more time where $i = 3$.

Example 2.4.19. Consider $V = \mathbb{R}^4$ with the standard inner product. Let U be the subspace spanned by the three vectors

$$\begin{bmatrix}1\\1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\-2\\0\\0\end{bmatrix} \text{ and } \begin{bmatrix}1\\0\\-1\\2\end{bmatrix}.$$

We will find an orthonormal basis for U. Firstly, by row reducing the matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 2 \end{bmatrix}$$

we see that it has rank 3. Therefore, these three vectors are linearly independent (So, they form a basis for U). Let us call them u_1, u_2, u_3 in order. The Gram-Schmidt process will produce us another basis $\{v_1, v_2, v_3\}$ of the same subspace U and this new basis will be orthonormal.

We start with computing $\|u_1\|$.

$$\|u_1\| = \sqrt{1^2 + 1^2 + 0^2 + 1^2} = \sqrt{3}.$$

Therefore, we put

$$v_1 = \frac{1}{\sqrt{3}} u_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}.$$

Next, we compute v_2. The Gram-Schmidt process tells us to put

$$\tilde{v}_2 = u_2 - \langle v_1, u_2 \rangle v_1.$$

So, we have

$$\begin{aligned} \tilde{v}_2 &= \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{\sqrt{3}} (1 \cdot 1 - 2 \cdot 1 + 0 + 0) \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} 4 \\ -5 \\ 0 \\ 1 \end{bmatrix}. \end{aligned}$$

We can compute that

$$\|\tilde{v}_2\| = \frac{1}{3}\sqrt{4^2 + 5^2 + 1^2} = \frac{1}{3}\sqrt{42}.$$

And therefore, we put

$$v_2 = \frac{1}{\frac{1}{3}\sqrt{42}} \frac{1}{3} \begin{bmatrix} 4 \\ -5 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{42}} \begin{bmatrix} 4 \\ -5 \\ 0 \\ 1 \end{bmatrix}.$$

Next, we compute $\tilde{v}_3$. The process tells us

$$\begin{aligned}
\tilde{v}_3 &= u_3 - \langle v_1, u_3\rangle v_1 - \langle v_2, u_3\rangle v_2 \\
&= \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix} - \frac{1}{\sqrt{3}}(1\cdot 1 + 0 + 0 + 1\cdot 2)\frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{42}}(4\cdot 1 + 0 + 0 + 1\cdot 2)\frac{1}{\sqrt{42}}\begin{bmatrix} 4 \\ -5 \\ 0 \\ 1 \end{bmatrix} \\
&= \frac{1}{7}\begin{bmatrix} -4 \\ -2 \\ -7 \\ 6 \end{bmatrix}.
\end{aligned}$$

We have

$$\|\tilde{v}_3\| = \frac{1}{7}\sqrt{16+4+49+36} = \frac{\sqrt{105}}{7}$$

and so

$$v_3 = \frac{7}{\sqrt{105}}\frac{1}{7}\begin{bmatrix} -4 \\ -2 \\ -7 \\ 6 \end{bmatrix} = \frac{1}{\sqrt{105}}\begin{bmatrix} -4 \\ -2 \\ -7 \\ 6 \end{bmatrix}.$$

2.5 Orthogonal projection

The Gram-Schmidt process gives us an orthonormal basis starting from an ordinary basis of a subspace. In the prood, we have seen algebraically why the process works: things cancel out! In this section, we will try to understand why it works in a more geometric way. The main theorem of this section will be the *Projektionsatz*, which we state below. Then, we will spend the rest of the section to prove it.

Theorem 2.5.1 (Projektionsatz1)**.** *Let U be a subspace of a Euklidean or unitary space V. Assume that* $\dim U = k$ *with a basis $\{b_1, \dots, b_k\}$. Then,*

1. *any vector $x \in V$ can be written uniquly as*

$$x = u + w$$

where $u \in U$ and $w \in U^{\perp}$, and

2. *since u is an element of U, we can write it as a linear combination $u = \lambda_1 b_1 + \cdots + \lambda_k b_k$ and these λ_i's satisfy the equation*

$$\begin{bmatrix} \langle b_1, b_1\rangle & \cdots & \langle b_1, b_k\rangle \\ \vdots & \dots & \vdots \\ \langle b_k, b_1\rangle & \cdots & \langle b_k, b_k\rangle \end{bmatrix}\begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_k \end{bmatrix} = \begin{bmatrix} \langle b_1, x\rangle \\ \vdots \\ \langle b_k, x\rangle \end{bmatrix}.$$

We will start with looking at the square matrix appering in the theorem.

Definition 2.5.2. Let V be a Euklidean or unitary space and let $\langle v_1, \dots, v_n \rangle$ be n vectors in V. Then, the $n \times n$-matrix whose entry in the ith row and the jth column is given by $\langle v_i, v_j \rangle$ is called the *Gram matrix* of $v_1, \dots, v_n$.

The Gram matrix has some nice properties, especially when the vectors are linearly independent.

Definition 2.5.3. Let $K = \mathbb{R}$ or $\mathbb{C}$, $V = K^n$ and let A be an $n \times n$-matrix.

1. A is called *Hermitian* or *self-adjoint* if $A^* = A$. A real self-adjoint matrix is called *symmetric.*

2. Let A be hermitian or symmetric. If for every nonzero $x \in V$, one has $x^*Ax \geq 0$, we call A *positive semi-definite.*

3. Let A be hermitian or symmetric. If for every nonzero $x \in V$, one has $x^*Ax > 0$, we call A *positive definite.*

Here is a lemma which we will find useful later.

Lemma 2.5.4. *If A is hermitian and positive definite, then A is invertible.*

Proof. Let $A = A^*$ be positive definite and let $Ax = 0$. We will show that the only solution to this is $x = 0$ and this will show that A is invertible as A is a square matrix. By definition of a hermitian matrix, we would have to have $x^*Ax > 0$ if x was nonzero. On the other hand, we have $Ax = 0$ and we have $x^*Ax = x^*0 = 0$. Thus, we conclude x is NOT nonzero. Hence, $x = 0$. $\square$

Proposition 2.5.5. *Let V be a Euklidean or unitary space and G be the Gram matrix of $v_1, \dots, v_n \in V$.*

1. G is hermitian and positive semi-definite.

2. G is positive definite if and only if $v_1, \dots, v_n$ are linearly independent.

Proof. The fact that G is hermitian follows directly from the definition of an inner product. Specifically, it follows from the axiom that for any $x, y \in V$, one must have $\langle x, y \rangle = \overline{\langle y, x \rangle}$. Let us now show that it is positive semi-definite. Take

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in K^n.$$

We have

$$\begin{aligned}
x^*Gx &= \begin{bmatrix} \overline{x_1} & \cdots & \overline{x_n} \end{bmatrix} \begin{bmatrix} \langle v_1, v_1\rangle & \cdots & \langle v_1, v_n\rangle \\ \vdots & \cdots & \vdots \\ \langle v_n, v_n\rangle & \cdots & \langle v_n, v_n\rangle \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\
&= \begin{bmatrix} \overline{x_1} & \cdots & \overline{x_n} \end{bmatrix} \begin{bmatrix} \sum_{j=1}^n \langle v_1, v_j\rangle x_j \\ \vdots \\ \sum_{j=1}^n \langle v_n, v_j\rangle x_j \end{bmatrix} \\
&= \sum_{i=1}^n \overline{x_i} \sum_{j=1}^n \langle v_i, v_j\rangle x_j \\
&= \sum_{i=1}^n \sum_{j=1}^n \overline{x_i} x_j \langle v_i, v_j\rangle \\
&= \sum_{i=1}^n \sum_{j=1}^n \langle x_i v_i, x_j v_j\rangle \\
&= \langle \sum_{i=1}^n x_i v_i, \sum_{j=1}^n x_j v_j\rangle
\end{aligned}$$

Now, since

$$\sum_{i=1}^n x_i v_i \text{ and } \sum_{j=1}^n x_j v_j$$

are literally the same thing, we see that x^*Gx equals taking the inner product of a vector with itself and therefore must be nonnegative.

Now, assume that $v_1, \ldots, v_n$ are linearly independent. Then, for $x \neq 0$, we must have $x_1 v_1 + \cdots + x_n v_n \neq 0$ by linear independence. Hence, the inner product we compute by x^*Gx can not be zero.

And now assume that $v_1, \ldots, v_n$ are linearly dependent. Then, there is at least one $y \neq 0$ such that $y_1 v_1 + \cdots + y_n v_n = 0$ (since $y \neq 0$, at least one of the y_i's must be nonzero and this is the definition of linear dependence). For this y, we would have

$$y^*Gy = \langle y_1 v_1 + \cdots + y_n v_n, y_1 v_1 + \cdots + y_n v_n\rangle = \langle 0, 0\rangle = 0.$$

Thus, linearly dependent vectors would give Gram matrices which are not positive definite. This means that positive definite Gram matrices must come from linearly independent vectors. □

Exercise 2.5.6. The proof above uses several properties of inner products. Go over the proof one more time with $n = 3$ and write at each step which property we used.

Proof of Projektionsatz. We are now ready to prove the main theorem of this section. Let $x \in V$ and $u \in U$ be arbitrary vectors. Since u is an element of U, we know that we can write $u = \lambda_1 b_1 + \cdots + \lambda_k b_k$ for some λ_i's. We would like to choose these λ_i's in such a way that $x - u$ belongs to $U^\perp$. We note that

$$\begin{aligned} x - u \in U^\perp &\iff \langle z, x-u\rangle = 0 \text{ for all } z \in U \\ &\iff \langle b_i, x-u\rangle = 0 \text{ for all } i = 1,\dots,k \\ &\iff \langle b_i, x\rangle - \langle b_i, u\rangle = 0 \text{ for all } i = 1,\dots,k \\ &\iff \langle b_i, x\rangle = \langle b_i, u\rangle \text{ for all } i = 1,\dots,k \\ &\iff \langle b_i, x\rangle = \langle b_i, \lambda_1 b_1 + \cdots + \lambda_k b_k\rangle \text{ for all } i = 1,\dots,k \\ &\iff \langle b_i, x\rangle = \lambda_1\langle b_i, b_1\rangle + \cdots + \lambda_k\langle b_i, b_k\rangle \text{ for all } i = 1,\dots,k. \end{aligned}$$

But now, if you look at this carefully, this is really telling us

$$x - u \in U^\perp \iff \begin{bmatrix} \langle b_1, b_1\rangle & \cdots & \langle b_1, b_k\rangle \\ \vdots & \cdots & \vdots \\ \langle b_k, b_1\rangle & \cdots & \langle b_k, b_k\rangle \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_k \end{bmatrix} = \begin{bmatrix} \langle b_1, x\rangle \\ \vdots \\ \langle b_k, x\rangle \end{bmatrix}.$$

Now, the $k \times k$ matrix here is the Gram matrix of the basis vectors $b_1, \dots, b_k$. Therefore, it is hermitian and positive definite by Proposition 2.5.5. Therefore, it is invertible by Lemma 2.5.4. Hence there exists a unique solution to this and therefore there exists a unique $u \in U$ such that $x = u + w$ with $u \in U$ and $w \in U^\perp$. □

Corollary 2.5.7. *Let U, V be as in Theorem 2.5.1. Then, we have*

1. $V = U \oplus U^\perp$.
2. $(U^\perp)^\perp = U$.
3. *If* $\dim V < \infty$, *then we have* $\dim V = \dim U + \dim U^\perp$.

Now that we proved the existence of a unique vector in Theorem 2.5.1, we should give it a name.

Definition 2.5.8. Let U be a finite dimensional subspace of a Euklidean or unitary space V. The *orthogonal projection* of V onto U is the function $\pi\colon V \to V$ defined by the following formula: for any $x \in V$, $\pi(x) = u$ where $x = u + w$ with $u \in U$ and $w \in U^\perp$ as in the Projektionsatz.

Exercise 2.5.9. Let U, V be as in Theorem 2.5.1 and $\pi\colon V \to V$ be the orthogonal projection of V onto U. Show the following:

1. π is a linear transformation.
2. The restriction of π on U is identity on U. That is, $\pi_{|U} = \mathrm{id}_U$.
3. $\mathrm{Im}(\pi) = U$ and $\ker(\pi) = U^\perp$.
4. $\pi \circ \pi = \pi$.

Example 2.5.10. Let $V = \mathbb{R}^4$ with the standard inner product and U be the subspace spanned by the vectors

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} \text{ and } u_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

For a given $x \in \mathbb{R}^4$, let us determine the orthogonal projection $\pi\colon V \to V$ of V onto U. We will apply Theorem 2.5.1. Computing the inner products, we get that the Gram matrix G of u_1, u_2 is

$$G = \begin{bmatrix} 6 & 3 \\ 3 & 3 \end{bmatrix}.$$

Moreover, for

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix},$$

we have $\langle u_1, x \rangle = x_1 - 2x_3 - x_4$ and $\langle u_2, x \rangle = x_1 + x_2 - x_3$. Therefore, the Projectionsatz tells us that the equation that we need to solve is

$$\begin{bmatrix} 6 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} x_1 - 2x_3 - x_4 \\ x_1 + x_2 - x_3 \end{bmatrix}.$$

By Lemma 2.5.4, we know that our 2×2 matrix is invertible. We can show that

$$\begin{bmatrix} 6 & 3 \\ 3 & 3 \end{bmatrix}^{-1} = \frac{1}{9} \begin{bmatrix} 3 & -3 \\ -3 & 6 \end{bmatrix}.$$

Hence, we get

$$\begin{aligned} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} &= \frac{1}{9} \begin{bmatrix} 3 & -3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} x_1 - 2x_3 - x_4 \\ x_1 + x_2 - x_3 \end{bmatrix} \\ &= \frac{-1}{3} \begin{bmatrix} x_2 + x_3 + x_4 \\ -x_1 - 2x_2 - x_4 \end{bmatrix}. \end{aligned}$$

Therefore, we have

$$\begin{aligned} \pi\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}\right) &= \frac{-1}{3}(x_2 + x_3 + x_4) \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} + \frac{1}{3}(x_1 + 2x_2 + x_4) \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} x_1 + 2x_2 - x_3 \\ x_1 + 2x_2 + x_4 \\ -x_1 + 2x_3 + x_4 \\ x_2 + x_3 + x_4 \end{bmatrix}. \end{aligned}$$

We started this section by saying our goal is to understand the Gram-Schmidt process more geometrically. The following exercises aim to do this.

Exercise 2.5.11. Let U, V be as in the Projektionsatz and assume that $\{b_1, \ldots, b_k\}$ is an *orthonormal basis* of U. Let $x \in V$.

1. Show that

$$\pi(x) = \langle b_1, x\rangle b_1 + \cdots + \langle b_k, x\rangle b_k.$$

2. Conclude that

$$x - \langle b_1, x\rangle b_1 - \cdots - \langle b_k, x\rangle b_k \in U^\perp.$$

Hint: What is the Gram matrix?

Exercise 2.5.12. Use the previous exercise to explain the Gram-Schmidt process.

Let us now continue with some other geometric ideas. Recall that in Theorem 2.4.2, we proved a version of the Pythagoras Theorem. We use this in the next theorem.

Theorem 2.5.13. *Let V be a Euklidean vectorspace and U a finite dimensional subspace of V. Let π be the orthogonal projection of V onto U and $x \in V$. Then, for any $u \in U$ with $u \neq \pi(x)$, we have*

$$\|x - \pi(x)\| < \|x - u\|.$$

That is, $\pi(x)$ is the closest *vector to x in U.*

Proof. We know that $x - \pi(x)$ is in $U^\perp$. On the other hand, we know that u and $\pi(x)$ are in U so that $\pi(x) - u \in U$ for any $u \in U$ as U is a subspace. This gives us that $x - \pi(x)$ is orthogonal to $\pi(x) - u$ for any $u \in U$. Then, by Theorem 2.4.2, we get

$$\|x - \pi(x)\|^2 + \|\pi(x) - u\|^2 = \|x - \pi(x) + \pi(x) - u\|^2 = \|x - u\|^2.$$

Since $\pi(x) \neq u$, we know that $\pi(x) - u \neq 0$ and thus, $\|\pi(x) - u\| \neq 0$. Therefore, we get $\|x - \pi(x)\|^2 < \|x - u\|^2$ which implies the desired inequality

$$\|x - \pi(x)\| < \|x - u\|.$$

□

This has some very important computational consequences. Assume that we have an $m \times n$ matrix A and $b \in \mathbb{R}^m$ such that the equation $Ax = b$ does not have a solution. But maybe we are not interested in an *exact* solution! Maybe, for our purposes, it is enough to find an x such that Ax is *close* to b. Let us rewrite this.

We have

$$Ax = b \iff b - Ax = 0 \iff \|b - Ax\| = 0.$$

If the equation $Ax = b$ is not solvable, we will never get an x with $\|b - Ax\| = 0$. But maybe we can try to minimize the quantity $\|b - Ax\|$. How can we do that? Well, before trying to attack the problem, let us rewrite it. We would like to find an $\hat{x} \in \mathbb{R}^n$ such that

$$\|b - A\hat{x}\| = \min_{x \in \mathbb{R}^m} \|b - Ax\|.$$

But wait, we can still rewrite this! Note that a vector is of the form Ax if and only if it belongs to the *image* of A. Hence, we have

$$\min_{x \in \mathbb{R}^m} \|b - Ax\| = \min_{u \in \mathrm{im}(A)} \|b - u\|.$$

Now we are getting closer. Let $V = \mathbb{R}^m$ and π be the orthogonal projection of V onto $\mathrm{im}(A)$. Then, by Theorem 2.5.13, we have

$$\|b - \pi(b)\| < \|b - u\|$$

for any $u \in \mathrm{im}(A)$ with $u \neq \pi(b)$. Hence, the $\hat{x}$ we are looking for is a vector which gives $A\hat{x} = \pi(b)$ where π is the orthogonal projection onto $\mathrm{im}(A)$. The following exercise revisits this entire discussion.

Exercise 2.5.14. Let A be an $m \times n$ matrix and $b \in \mathbb{R}^m$. Let π be the orthogonal projection of $\mathbb{R}^m$ onto $\mathrm{im}(A)$.

1. Show that $b - \pi(b)$ is orthogonal to Ax for any $x \in \mathbb{R}^n$.
2. Show that $(b - \pi(b))^T Ax = 0$ for any $x \in \mathbb{R}^n$.
3. Show that $(A^T(b - \pi(b)))^T x = 0$ for any $x \in \mathbb{R}^n$.
4. Show that $x \in \ker((A^T(b - \pi(b)))^T)$ for any $x \in \mathbb{R}^n$.
5. Show that $(A^T(b - \pi(b)))^T$ is the zero matrix/zero vector.
6. Show that $A^T(b - \pi(b))$ is the zero matrix/zero vector.
7. Let $\hat{x}$ be as in the above discussion. Show that $A^T(b - A\hat{x})$ is zero.
8. Let $\hat{x}$ be as in the above discussion. Show that $A^T b = A^T A\hat{x}$.

The exercise consists of one sentence proofs. But if you can solve all of them you proved the following.

Theorem 2.5.15. *Let A be an $m \times n$ matrix. Then, $\hat{x} \in \mathbb{R}^n$ satisfies*

$$\|b - A\hat{x}\| = \min_{x \in \mathbb{R}^n} \|b - Ax\|$$

if and only if $\hat{x}$ is a solution to the system

$$A^T Ax = A^T b.$$

2.6 Adjoint functions

In the last section of this chapter, we will talk about adjoint functions. Let V and W be two Euklidean or unitary spaces and $f\colon V \to W$ be a linear transformation. For a vector $v \in V$ and $w \in W$, the quantity $\langle v, w\rangle$ does not make any sense. Because these two vectors live in different places, we can not take their inner product. However, we can send v to W via f and then take the inner product in W! More precisely, the quantity

$$\langle f(v), w\rangle_W$$

makes sense where $\langle -, -\rangle_W$ denotes the inner product in W. What we want in this section is to do this the other way around. Instead of sending v to W, send w to V somehow and take the inner product inside V. Can we do this? Let us rewrite this:

Question 2.6.1. *Let V and W be two Euklidean or unitary spaces and $f\colon V \to W$ be a linear transformation. Can we find a function g such that we have*

$$\langle f(v), w\rangle_W = \langle v, g(w)\rangle_V \tag{2.6.1.1}$$

for any $v \in V$ and $w \in W$ where $\langle -, -\rangle_W$ denotes the inner product in W and $\langle -, -\rangle_V$ denotes that in V?

Remark 2.6.2. It is important to understand that this is not an inverse of f.

Remark 2.6.3. Assume that the equation in 2.6.1.1 holds. Then, we have

$$\langle w, f(v)\rangle = \overline{\langle f(v), w\rangle} = \overline{\langle v, g(w)\rangle} = \langle g(w), v\rangle.$$

The question has an affirmative answer. We are stating this as a theorem.

Theorem 2.6.4. *Let $f\colon V \to W$ be a linear transformation between finite dimensional vector spaces and $\mathsf{B} = \{v_1, \ldots, v_n\}$ be an orthonormal basis of V. Consider the function $g\colon W \to V$ given by the rule*

$$g(w) = \langle f(v_1), w\rangle_W v_1 + \cdots + \langle f(v_n), w\rangle_W v_n.$$

Then, g satisfies the equality 2.6.1.1.

Proof. Firstly, since an inner product is linear in the second component, g is a linear transformation. Now, given $w \in W$ and $v = a_1v_1 + \cdots + a_nv_n$. Put $\ell_i = \langle f(v_i), w\rangle$. Then,

$$\begin{aligned}\langle f(v), w\rangle &= \langle f(a_1v_1 + \cdots + a_nv_n), w\rangle \\ &= \overline{a_1}\langle f(v_1), w\rangle + \cdots + \overline{a_n}\langle f(v_n), w\rangle \\ &= \overline{a_1}\ell_1 + \cdots + \overline{a_n}\ell_n.\end{aligned}$$

On the other hand, we have

$$\langle v, g(w)\rangle = \langle v, \ell_1v_1 + \cdots \ell_nv_n\rangle = \ell_1\langle v, v_1\rangle + \cdots + \ell_n\langle v, v_n\rangle.$$

And at the same time, we have

$$\langle v, v_i\rangle = \langle a_1 v_1 + \cdots + a_n v_n, v_i\rangle = \overline{a_1}\langle v_1, v_i\rangle + \cdots + \overline{a_n}\langle v_n, v_i\rangle = \overline{a_i}$$

since our basis is an orthonormal basis. Combining this, we get

$$\langle v, g(w)\rangle = \ell_1\overline{a_1} + \cdots + \ell_n\overline{a_n}.$$

This shows us the desired equality since multiplication of scalars is commutative. $\square$

Lemma 2.6.5. *Let $f\colon V \to W$ be as in Question 2.6.1. If $g, h\colon W \to V$ are functions that satisfy equality 2.6.1.1, then we must have $g = h$.*

Proof. For any $v \in V$ and $w \in W$, we have

$$\langle v, g(w)\rangle = \langle f(v), w\rangle = \langle v, h(w)\rangle.$$

This gives us

$$\langle v, g(w)\rangle = \langle v, h(w)\rangle$$

which means

$$\langle v, h(w) - g(w)\rangle = 0.$$

Since this equality is true for all v and w, we conclude that $h(w) - g(w) = 0$ which finishes the proof. $\square$

We have proved existence and we have proved uniqueness. Now, we give a name.

Definition 2.6.6. Let $f\colon V \to W$ be as in Question 2.6.1. The unique function from W to V which satisfies the equality 2.6.1.1 is called the *adjoint function* of f and is denoted by f^*.

Example 2.6.7. Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ be given by the rule

$$f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ 2x_1 + x_2 \\ -x_2 \end{bmatrix}$$

Let's find f^*. Following the existence theorem, we start with choosing an orthonormal basis for $\mathbb{R}^2$. We will choose it to be the standard basis $\{e_1, e_2\}$. We have

$$f(e_1) = f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$f(e_2) = f\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

and therefore,

$$
\begin{aligned}
f^*\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) &= \langle \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rangle \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \langle \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rangle \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
&= (x+2y) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (y-z) \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} x+2y \\ y-z \end{bmatrix}
\end{aligned}
$$

Example 2.6.8. Now, let V be the space of $n \times n$ matrices with entries in $\mathbb{C}$, together with the inner product $\langle A, B\rangle := \operatorname{tr}(A^*B)$. Now, fix an $n \times n$ matrix M. Consider the transformation $\phi_M \colon V \to V$ given by $\phi_M(A) = MA$. We have

$$\langle \phi_M(A), B\rangle = \langle MA, B\rangle = \operatorname{tr}((MA)^*B) = \operatorname{tr}((A^*M^*)B) = \operatorname{tr}(A^*(M^*B)) = \langle A, M^*B\rangle$$

for any $A, B \in V$. Hence, we have

$$\phi_M^*(X) = M^*X$$

and so $\phi_M^* = \phi_{M^*}$.

The following properties are standard exercises.

Exercise 2.6.9. Let $f, \phi \colon U \to V$ and $g \colon V \to W$ be linear transformations between Euklidean (or unitary) spaces. Assume that λ is a scalar. Then, we have

1. $(f+\phi)^* = f^* + \phi^*$.
2. $(\lambda f)^* = \overline{\lambda} f^*$.
3. $(g \circ f)^* = f^* \circ g^*$.
4. $(f^*)^* = f$.

We have used the notation A^* before to denote the conjugate transpose of a matrix with complex entries. We are also using the notation f^* here. Does this suggest a relationship? This is a very natural question. One can make it more precise by asking the following.

Question 2.6.10. *Let V, W be two finite dimensional Euklidean or unitary spaces and $f \colon V \to W$ be linear. Assume that B is a basis of V and C is a basis for W. Then, is it true that we have an equality*

$$[f^*]_{\mathsf{C},\mathsf{B}} = [f]^*_{\mathsf{B},\mathsf{C}} \text{ ?}$$

Unfortunately, the answer is not a yes. You will see an exercise verifying this. But, there is one (and a very important one) case where this equality is true.

Theorem 2.6.11. *Let V, W be two finite dimensional Euklidean or unitary spaces and $f: V \to W$ be linear. Assume that B is an orthonormal basis of V and C is an orthonormal basis for W. Then, we have an equality*

$$[f^*]_{\mathsf{C},\mathsf{B}} = [f]^*_{\mathsf{B},\mathsf{C}}.$$

Proof. Put $\mathsf{B} = \{v_1, \ldots, v_n\}$ and $\mathsf{C} = \{w_1, \ldots, w_m\}$. We know from the first chapter that $[f]_{\mathsf{B},\mathsf{C}} = (a_{ij}) \in M_{m\times n}(K)$ with $f(v_i) = \sum_{k=1}^m a_{ki} w_k$ for $i = 1, \ldots, n$ and $[f^*]_{\mathsf{C},\mathsf{B}} = (b_{ij}) \in M_{n\times m}(K)$ with $f^*(w_j) = \sum_{k=1}^n b_{kj} v_k$ for $j = 1, \ldots, m$. This is just how one writes the matrix of a linear transformation with given bases.

The adjoint function requires the equality 2.6.1.1, let's use that to see what we can get. We have

$$\langle w_j, f(v_i)\rangle = \langle w_j, \sum_{k=1}^m a_{ki} w_k\rangle = \sum_{k=1}^m a_{ki}\langle w_j, w_k\rangle = a_{ji}$$

where the first equality is by the above paragraph, the second using linearity in the second component and the third by the fact that C is an orthonormal basis: when $j \neq k$, we have $\langle w_j, w_k\rangle = 0$ and $\langle w_j, w_j\rangle = 1$. Next, we have

$$\langle f^*(w_j), v_i\rangle = \langle \sum_{k=1}^n b_{kj} v_k, v_i\rangle = \sum_{k=1}^n \overline{b_{kj}}\langle v_k, v_i\rangle = \overline{b_{ij}}.$$

Equality 2.6.1.1 tells us that we must have $a_{ji} = \overline{b_{ij}}$ and this is exactly what we are supposed to prove. □

Theorem 2.6.12. *Let $f: V \to W$ be a linear map between finite dimensional Euklidean or unitary spaces. Then, we have*

1. $\ker f = (\operatorname{im}(f^*))^\perp$.
2. $\ker(f^*) = (\operatorname{im}(f))^\perp$.
3. $V = \ker(f) \oplus \operatorname{im}(f^*)$.
4. $W = \ker(f^*) \oplus \operatorname{im}(f)$.

Proof. We will prove the first two and leave the last two statements as an exercise. In order to show the equality $\ker f = (\operatorname{im}(f^*))^\perp$, we must show the inclusions $\ker f \subseteq (\operatorname{im}(f^*))^\perp$ and $(\operatorname{im}(f^*))^\perp \subseteq \ker f$. We will start with the first one.

Let $x \in \ker(f)$. This means that $f(x) = 0$. We should show that $x \in (\operatorname{im}(f^*))^\perp$. In order to do this, we pick an arbitrary element $y \in \operatorname{im}(f^*)$. We should show that x is orthogonal to y. Since $y \in \operatorname{im}(f^*)$, there must be a $w \in W$ such that $y = f^*(w)$. We have

$$\langle x, y\rangle = \langle x, f^*(w)\rangle = \langle f(x), w\rangle = \langle 0, w\rangle = 0.$$

As explained, this shows the inclusion $\ker f \subseteq (\mathrm{im}(f^*))^\perp$. For the other inclusion, we pick an element $x \in (\mathrm{im}(f^*))^\perp$. We want to show that $f(x) = 0$. We have

$$f(x) = 0 \iff \|f(x)\| = 0 \iff \langle f(x), f(x)\rangle = 0 \iff \langle x, f^* f(x)\rangle = 0.$$

Since we picked $x \in (\mathrm{im}(f^*))^\perp$ and since $f^* f(x) \in \mathrm{im}(f^*)$, we have $\langle x, f^* f(x)\rangle = 0$ and so we have $f(x) = 0$ as required. We showed the first statement.

In order to show the second statement, we use the first statement:

$$\ker(f^*) = (\mathrm{im}((f^*)^*)^\perp = (\mathrm{im}(f)^\perp.$$

□

Exercise 2.6.13. The last two statements of the previous theorem follow easily from a theorem that we proved earlier. Which theorem is it?

Chapter 3

Determinants

In this section, we are going to talk about a numerical invariant of a linear transformation/matrix which has plenty of applications. Our plan is to start talking about the determinants and their important properties without rigorous proofs at the beginning. We will slowly introduce more proofs throughout the chapter.

3.1 First properties

Determinant of a square matrix with entries in a field k is a scalar in k. That is, determinant defines a function from the space of $n \times n$ matrices into k. However, there are actually several ways to think about a square matrix. Consider the matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}.$$

1. We can see this matrix as one single 2×2 matrix.
2. We can see it consisting of two columns
$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} \text{ and } \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$
3. We can see it consisting of two rows
$$\begin{bmatrix} 1 & 2 \end{bmatrix} \text{ and } \begin{bmatrix} 3 & 5 \end{bmatrix}$$
4. Or we can just think it as four scalars $1, 2, 3$ and 5.

In this chapter, we will be focusing on the first two ways of considering a matrix. We will define the determinant rigorously seeing it as a set of columns as in the second case above, for instance.

Determinant is a tool which can tell us important things about a square matrix. For instance, we can use it to check whether our matrix or linear transformation is invertible. This will be very handy in the next chapter when we talk about the characteristic polynomial of a matrix, for example.

The precise relationship between the determinant of a matrix and its invertibility is as follows.

Theorem 3.1.1. *An $n \times n$ matrix is invertible if and only if its determinant is not zero.*

Let us prove this theorem using some facts that we have not yet proved but will prove in this chapter. Here are three things that you need to know about determinants in order to prove the theorem:

1. The determinant of the identity matrix is 1.
2. If one of the columns of a matrix is only zeroes, then the determinant of that matrix is zero.
3. The determinant is multiplicative: if A and B are two $n \times n$ matrices, then we have

$$\det(AB) = \det A \cdot \det B.$$

Proof of Theorem 3.1.1. Let A be an $n \times n$ matrix with entries in a field k. Assume that it is invertible so that there exists a matrix B such that $AB = I$ where I is the $n \times n$ identity matrix. Then, by the first and the last properties above, we have

$$\det(AB) = \det(I) = 1 \text{ and } \det(AB) = \det(A)\det(B).$$

This gives us

$$\det(A)\det(B) = 1.$$

This tells us that the determinant of A can not be zero. This shows one direction.

For the other direction, we want to show that if the determinant is nonzero, then the matrix is invertible. Instead, we will prove the contrapositive: if the matrix is not invertible, then the determinant is zero. For this, assume that A is not invertible. Then, there exists a nonzero vector $v \in k^n$ such that $Av = 0$. Now, since v is not the zero vector, there exists a basis of k^n containing v. Consider the matrix C whose columns are these basis vectors. Assume for the ease of the proof that the first column of C is v. Since the columns of C are linearly independent, we know that C is invertible. So we already know that $\det(C) \neq 0$. Now consider the product AC. We have

$$\det(AC) = \det(A)\det(C).$$

And the first column of AC is given by Av. But this means that the first column of AC is zero. So, by the second property above, we know that $\det(AC) = 0$. Hence, we have

$$\det(A)\det(C) = 0.$$

Since $\det(C) \neq 0$, we must have $\det(A) = 0$. $\square$

3.2 Determinant for small matrices

Before defining the determinant of a matrix rigorously, we are going to see examples on matrices of small size. That is, we will look at the case of 1×1, 2×2 and 3×3 matrices.

A 1×1 matrix is nothing but a scalar. The determinant of this scalar is just that scalar. For instance, if $k=\mathbb{R}$ and we are looking at the 1×1 matrix $[2]$, then we have $\det([2])=2$.

The case of 2×2 matrices is richer. Given a matrix

$$A=\begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant of A is given by $\det(A)=ad-bc$. This is in fact the area of the parallellogram spanned by the column vectors of A (depending on the positions of these vectors, could also be the negative of the area). We have shown this in class, with a beautiful picture Özgür drew on the board.

Exercise 3.2.1. Show that the area of the parallellogram spanned by the vectors

$$\begin{bmatrix} a \\ c \end{bmatrix} \text{ and } \begin{bmatrix} b \\ d \end{bmatrix}$$

is $\pm(ad-bc)$.

We can see the properties that we described above directly by computation in the case of a 2×2 matrix.

Exercise 3.2.2. Show that

$$\det\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}=1.$$

Exercise 3.2.3. Show that

$$\det\begin{bmatrix} 0 & x \\ 0 & y \end{bmatrix}=0.$$

Exercise 3.2.4. Let

$$A=\begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } X=\begin{bmatrix} x & y \\ z & t \end{bmatrix}.$$

1. Compute AX.
2. Show that $\det(AX)=\det(A)\det(X)$ by direct computation.

Exercise 3.2.5. Show that if A is a 2×2 matrix, then $\det(A)=\det(A^T)$.

Exercise 3.2.6. Show that

$$\det\begin{bmatrix} 2a & b \\ 2c & d \end{bmatrix}=2\cdot\det\begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ and}$$
$$\det\begin{bmatrix} 2a & 2b \\ c & d \end{bmatrix}=2\cdot\det\begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Exercise 3.2.7. Show that

$$\det \begin{bmatrix} a & b+x \\ c & d+y \end{bmatrix} = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \det \begin{bmatrix} a & x \\ c & y \end{bmatrix}.$$

Confirm this by choosing numbers instead of a, b, c, d, x, y, z, t.

Exercise 3.2.8. Show that

$$\det \begin{bmatrix} b & a \\ d & c \end{bmatrix} = -\det \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

.

Exercise 3.2.9. Show that

$$\det \begin{bmatrix} p & p \\ q & q \end{bmatrix} = 0$$

by direct computation.

We now continue with the case of 3×3 matrices: the following is known as the *Sarrus* rule for the determinant of 3×3 matrices. Be careful: it only works for the 3×3 case.

$$\det \begin{bmatrix} a & b & c \\ x & y & z \\ p & q & t \end{bmatrix} = ayt + bzp + cxq - cyp - bxt - azq.$$

Exercise 3.2.10. Let A be a 3×3 matrix. By using Sarrus Rule, show the following.

1. $\det(A) = \det(A^T)$.
2. $\det(I_{3\times 3}) = 1$.
3. $\det(A) = 0$ if A has two identical columns.
4. $\det(A) = 0$ if A has two identical rows.

3.3 The definition

We are now ready to give the definition of *a* determinant function. Here, note that we said *a determinant function* and not *the determinant function*. It will not be clear from our definition, at first, why such a function should be unique. But we will see later that there is only such function which satisfies the properties of a determinant. Recall the discussion at the beginning of this section regarding different ways of seeing a matrix. In this definition, we will see a matrix as a combination of its columns.

Definition 3.3.1. Let k be a field. A function $D \colon k^n \times k^n \to k$ is called a *determinant function* on the space of $n \times n$ matrices with entries from k if it satisfies the following properties.

1. D is multilinear, which means that we have

$$D(v_1, \ldots, v_i + w_i, \ldots, v_n) = D(v_1, \ldots, v_i, \ldots, v_n) + D(v_1, \ldots, w_i, \ldots, v_n)$$
$$D(v_1, \ldots, \lambda v_i, \ldots, v_n) = \lambda D(v_1, \ldots, v_i, \ldots, v_n)$$

2. If $v_i = v_j$ for some $i \neq j$, (that is if two columns are identical), then

$$D(v_1, \ldots, v_n) = 0.$$

3. The determinant of the identity matrix is 1. That is, if $\{e_1, \ldots, e_n\}$ denotes the standard basis for k^n, then we have

$$D(e_1, \ldots, e_n) = 0.$$

Remark 3.3.2. The first property does NOT tell you that $\det(A + B) = \det(A) + \det(B)$. In fact, this equality is usually wrong.

Remark 3.3.3. You have already proved in exercises in the previous section that the function

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto ad - bc$$

is a determinant function for 2×2 matrices.

Definition 3.3.4. Let $n \geq 2$ and A be an $n \times n$ matrix with entries from a field k. Denote by A^{ij} the $(n-1) \times (n-1)$ matrix we obtain from A by deleting row i and column j.

Example 3.3.5. Let

$$A = \begin{bmatrix} 1 & 5 & 6 \\ 3 & 2 & 4 \\ 1 & 8 & 0 \end{bmatrix}.$$

Then, by colouring the first row and the second column of A as

$$A = \begin{bmatrix} \color{red}{1} & \color{red}{5} & \color{red}{6} \\ 3 & \color{red}{2} & 4 \\ 1 & \color{red}{8} & 0 \end{bmatrix},$$

we see that

$$A^{12} = \begin{bmatrix} 3 & 4 \\ 1 & 0 \end{bmatrix}.$$

Definition 3.3.6. Let A be an $n \times n$ matrix and $i \in \{1, \ldots, n\}$. Then, we define

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(A^{ij}).$$

There are two things that we need to show in order for this definition to make sense:

1. It gives us the same number no matter which i we choose. Because our definition chooses one i and fixes it.
2. This definition actually satisfies the properties of a determinant function as above.

Exercise 3.3.7. Show that the function given in Definition 3.3.6 satisfies the properties of a determinant given in Definition 3.3.1. Hint: proof is by induction.

Example 3.3.8. Let us compute $\det(A)$ where

$$A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 2 & 2 & 1 & 0 \\ 2 & 3 & 1 & 2 \\ 3 & 2 & -1 & 4 \end{bmatrix}.$$

We will compute the determinant using the first row. We have

$$A^{11} = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 1 & 2 \\ 2 & -1 & 4 \end{bmatrix}, A^{12} = \begin{bmatrix} 2 & 1 & 0 \\ 2 & 1 & 2 \\ 3 & -1 & 4 \end{bmatrix}, A^{13} = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 3 & 2 \\ 3 & 2 & 4 \end{bmatrix}, A^{14} = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 3 & 1 \\ 3 & 2 & -1 \end{bmatrix}$$

and

$$\det(A) = 1 \cdot \det(A^{11}) - 1 \cdot \det(A^{12}) + 2\det(A^{13}) - 3\det(A^{14}).$$

We broke the 4×4 determinant computation into 4 computations of 3×3 determinants. We have

$$\begin{aligned}
\det(A^{11}) &= 2\det\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} - \det\begin{bmatrix} 3 & 2 \\ 2 & 4 \end{bmatrix} = 2 \cdot 6 - 8 = 4 \\
\det(A^{12}) &= 2\det\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} - \det\begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix} = 2 \cdot 6 - 2 = 10 \\
\det(A^{13}) &= 2\det\begin{bmatrix} 3 & 2 \\ 2 & 4 \end{bmatrix} - 2\det\begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix} = 2 \cdot 8 - 2 \cdot 2 = 12 \\
\det(A^{14}) &= 2\det\begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix} - 2\det\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} + \det\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \\
&= 2 \cdot (-5) - 2 \cdot (-5) + (-5) = -5.
\end{aligned}$$

Hence,

$$\det(A) = 4 - 10 + 2 \cdot 12 - 3 \cdot (-5) = 33.$$

Theorem 3.3.9. *Let A' be the matrix obtained by swapping two columns of an $n \times n$ matrix A. Then we have that* $\det(A') = -\det(A)$.

Proof. We will show the idea of the proof in the 3×3 case and leave the general case as an exercise. Now consider

$$A = \begin{bmatrix} a & x & p \\ b & y & q \\ c & z & r \end{bmatrix} \text{ and } A' = \begin{bmatrix} x & a & p \\ y & b & q \\ z & c & r \end{bmatrix}.$$

We know, by the second property of a determinant in Definition 3.3.1, that

$$\det \begin{bmatrix} a+x & a+x & p \\ b+y & b+y & q \\ c+z & c+z & r \end{bmatrix} = 0.$$

On the other hand, by the first property, we know that

$$\begin{aligned}
\det \begin{bmatrix} a+x & a+x & p \\ b+y & b+y & q \\ c+z & c+z & r \end{bmatrix} &= \det \begin{bmatrix} a & a+x & p \\ b & b+y & q \\ c & c+z & r \end{bmatrix} + \det \begin{bmatrix} x & a+x & p \\ y & b+y & q \\ z & c+z & r \end{bmatrix} \\
&= \det \begin{bmatrix} a & a & p \\ b & b & q \\ c & c & r \end{bmatrix} + \det \begin{bmatrix} a & x & p \\ b & y & q \\ c & z & r \end{bmatrix} \\
&+ \det \begin{bmatrix} x & a & p \\ y & b & q \\ z & c & r \end{bmatrix} + \det \begin{bmatrix} x & x & p \\ y & y & q \\ z & z & r \end{bmatrix} \\
&= 0 + \det(A) + \det(A') + 0.
\end{aligned}$$

Combining, we get the desired result. □

Exercise 3.3.10. Prove the above theorem for a general $n \times n$ matrix. Hint: All you need to do is to follow the same idea by using general a_{ij} instead of a, b, c, x, y, z etc.

3.4 The Leibniz rule

We have defined a determinant function and gave an example of a function that satisfies these properties. The next obvious question is: "is there any other such function"? In this section, we will prove uniqueness. This will be via permutations of a set with n elements. In fact, for $n = 3$, we have already seen this earlier as the "Sarrus rule". We start with recalling some definitions.

Definition 3.4.1. Let $[n]\colon = \{1, \ldots, n\}$. A bijection $\sigma\colon [n] \to [n]$ is called a *permutation.* We write a permutation as

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}.$$

The set of all permutations forms a group under composition which is called the *symmetric group* on n elements. We denote this group by S_n.

Example 3.4.2. For $n = 2$, the symmetric group S_2 has two elements:

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

As a group, S_2 is isomorphic to $(\mathbb{Z}/2\mathbb{Z}, +)$.

Example 3.4.3. For $n = 3$, the symmetric group S_3 has six elements:

$$\begin{matrix} & & \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \\ \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, & \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, & \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \\ & \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, & \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \end{matrix}$$

Note that the permutation in the first row is the identity function, the permutations in the second row fix exactly one element and the permutations on the last row do not fix anything.

Exercise 3.4.4. Is S_n an abelian group for $n \geq 3$?

We now need another definition for permutations which are not identity but very close to being identity. What does this mean? If a permutation is not the identity transformation, it needs to move at least one element. But note that it is impossible to move exactly one element! One needs at least two elements to be moved, if the permutation is not the identity.

Definition 3.4.5. Let $i, j \in [n]$. A *transposition* $\tau := (ij)$ of $[n]$ is the non-identity permutation which fixes all the elements which are not i and j. This means we have

$$\tau(i) = j \quad , \quad \tau(j) = i \quad \text{and} \quad \tau(x) = x \text{ if } x \neq i, j.$$

Note that we can also write $\tau = (ij)$ as

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & i & \cdots & j & \cdots & n \\ 1 & 2 & \cdots & j & \cdots & i & \cdots & n \end{pmatrix}.$$

Exercise 3.4.6. Show that for any transposition τ on $[n]$, we have

1. $\tau \circ \tau = \text{id}$.
2. $\tau^{-1} = \tau$.

The nice thing about transpositions is that they are the simplest non-identity permutations. However, they are very powerful: we can write any permutation as a product (composition) of transpositions. This is similar to the fact that any positive integer can be written as a product of (powers) of prime numbers.

Theorem 3.4.7. *Every permutation in S_n (for $n \geq 2$) can be written as a product of transpositions.*

Proof. The proof is by induction. For $n = 2$, as we have seen, we have to elements:

$$\epsilon = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \sigma = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

Note that σ itself is a transposition, so we have $\sigma = \sigma$. And note that we can write $\epsilon = \sigma \circ \sigma$. So, all elements in S_2 can be written as a (possibly trivial) product of σ's. Now, assume that the statement is correct for all $i \leq n$. We will show that it is true for $n+1$. Given $\sigma \in S_{n+1}$. Consider the transposition $\tau \in S_{n+1}$ with $\tau(n+1) = j$. Since τ is a transposition, we have $\tau(j) = n+1$. And we have $\tau(x) = x$ for all $x \neq j, n+1$. We now consider the composition $\tau \circ \sigma$:

$$(\tau \circ \sigma)(n+1) = \tau(\sigma(n+1)) = \tau(j) = n+1.$$

This means that the composition $\tau \circ \sigma$ fixes $n+1$. And this means that $\tau \circ \sigma$ only permutes the elements from 1 to n. In other words, $\tau \circ \sigma$ is a permutation of $[n]$. By induction hypothesis, we can write $\tau \circ \sigma$ as a product of permutations of $[n]$:

$$\tau \circ \sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_\ell.$$

Multiplying both sides on the left by τ, by the previous exercise, we get

$$\begin{aligned} \tau \circ \tau \circ \sigma &= \tau \circ \tau_1 \circ \tau_2 \circ \cdots \circ \tau_\ell \\ \mathrm{id} \circ \sigma &= \tau \circ \tau_1 \circ \tau_2 \circ \cdots \circ \tau_\ell \\ \sigma &= \tau \circ \tau_1 \circ \tau_2 \circ \cdots \circ \tau_\ell \end{aligned}$$

as required. □

Example 3.4.8. Consider the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \in S_3.$$

By following the proof, we look at where the last element goes: $\sigma(3) = 1$. And we consider the transposition $\tau = (13)$. This means $\tau(1) = 3, \tau(2) = 2, \tau(3) = 1$. Therefore,

$$\begin{aligned} (\tau \circ \sigma)(1) &= \tau(\sigma(1)) = \tau(2) = 2 \\ (\tau \circ \sigma)(2) &= \tau(\sigma(2)) = \tau(3) = 1 \\ (\tau \circ \sigma)(3) &= \tau(\sigma(3)) = \tau(1) = 3 \end{aligned}$$

Thus, we see that $\tau \circ \sigma$ is the transposition $\tau_1 = (13)$: it fixes 2 and swaps 1 and 3. Thus, we get

$$\tau \circ \sigma = \tau_1$$

and multiplying both sides on the left by τ, we get

$$\sigma = \tau \circ \tau_1.$$

Above, we mentioned that one can compare the fact that we can write any permutation as a product of transpositions to the fact that we can write any positive integer as a product of primes. However, there is a very big difference between the two: factorisation into primes is unique! This does not work when we want to write a permutation as a product of transpositions. Above, you have seen that for any transposition τ one has $\tau^2 = \mathrm{id}$. Therefore, a product $\tau_1 \circ \tau_2$ is the same as $\tau_1 \circ \tau_2 \circ \tau_3 \circ \tau_3$ where τ_1, τ_2, τ_3 are transpositions, for instance. Let us record this as a remark.

Remark 3.4.9. Assume $n \geq 2$ and let $\sigma \in S_n$ be a permutation. The number of transpositions needed to write σ as a product of transpositions is not well-defined. That is, there may be two distinct numbers k, ℓ such that

$$\sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_k = \rho_1 \circ \rho_2 \circ \cdots \circ \rho_\ell$$

where τ_i's and ρ_j's are transpositions.

However, we still have something useful. While the number of transpositions needed is not well-defined, its parity is! That is, in the above remark we actually must have $k = \ell \mod 2$. Or in other words, $(-1)^k = (-1)^\ell$. We record this as a proposition.

Proposition 3.4.10. *Assume $n \geq 2$ and let $\sigma \in S_n$ be a permutation. Assume that we have*

$$\sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_k = \rho_1 \circ \rho_2 \circ \cdots \circ \rho_\ell.$$

Then, we have $(-1)^k = (-1)^\ell$.

Proof. Let I be the $n \times n$ identity matrix. Then S_n *acts* on I by permuting its columns. For any permutation $\pi \in S_n$, let us denote by πI the matrix whose columns are swapped according to π. Note that if τ is a transposition, then τI is obtained by I by swapping only two columns. Then, by Theorem 3.3.9, we get that $\det(I) = -\det(\tau I)$ for any transposition. Since $\det(I) = 1$ by definition, we see that $\det(\tau I) = -1$ for any transposition τ.

Now, letting σ as in the theorem, we see that

$$\begin{aligned} \det(\sigma) &= \det(\tau_1) \cdot \det(\tau_2) \cdot \cdots \cdot \det(\tau_k) = (-1)^k, \text{ and} \\ \det(\sigma) &= \det(\rho_1) \cdot \det(\rho_2) \cdot \cdots \cdot \det(\rho_\ell) = (-1)^\ell. \end{aligned}$$

This shows that $(-1)^k = (-1)^\ell$. □

This allows us to make the following definition.

Definition 3.4.11. Let $\sigma \in S_n$. The *signature* of σ, denoted $\mathrm{sgn}(\sigma)$, is the well-defined number $(-1)^k$ where there are transpositions $\tau_1, \ldots, \tau_k$ with

$$\sigma = \tau_1 \circ \tau_2 \circ \cdots \circ \tau_k.$$

Exercise 3.4.12. Let $n \geq 2$.

1. Show that for any transposition $\tau \in S_n$, one has $\mathrm{sgn}(\tau) = -1$.

2. Show that for any two permutations $\sigma_1, \sigma_2 \in S_n$, one has $\operatorname{sgn}(\sigma_1 \circ \sigma_2) = \operatorname{sgn}(\sigma_1) \cdot \operatorname{sgn}(\sigma_2)$.
3. Show that for any permutation $\sigma \in S_n$, one has $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$.

We are now ready to prove uniqueness of the determinant.

Theorem 3.4.13. *For each $n \geq 1$, there exists a unique function which satisfies the properties in Definition 3.3.1. This function satisfies the rule*

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n},$$

known as the Leibniz rule, where $A = (a_{i,j})$ is an $n \times n$ matrix.

Proof. We will illustrate the proof for the case $n = 4$ and leave the general case as an exercise. Let A_i denote the ith column of A, so that we have

$$A_i = \begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \\ a_{4i} \end{bmatrix} = \begin{bmatrix} a_{1i} \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ a_{2i} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ a_{3i} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ a_{4i} \end{bmatrix} = a_{1i}e_1 + a_{2i}e_2 + a_{3i}e_3 + a_{4i}e_4$$

where $\{e_1, \ldots, e_4\}$ is the standard basis for $\mathbb{R}^4$. In particular, for $i = 1$, we have

$$A_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{bmatrix} = a_{11}e_1 + a_{21}e_2 + a_{31}e_3 + a_{41}e_4.$$

Then, by the defining properties of a determinant as in Definition 3.3.1, we have

$$\begin{aligned} \det(A) &= \det[A_1, A_2, A_3, A_4] \\ &= \det[a_{11}e_1 + a_{21}e_2 + a_{31}e_3 + a_{41}e_4, A_2, A_3, A_4] \\ &= a_{11} \det[e_1, A_2, A_3, A_4] + a_{21} \det[e_2, A_2, A_3, A_4] \\ &\quad + a_{31} \det[e_3, A_2, A_3, A_4] + a_{41} \det[e_4, A_2, A_3, A_4]. \end{aligned}$$

Let us focus on the first summand $a_{11} \det[e_1, A_2, A_3, A_4]$. As above, write

$$A_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \\ a_{42} \end{bmatrix} = a_{12}e_1 + a_{22}e_2 + a_{32}e_3 + a_{42}e_4.$$

Then,

$$\begin{aligned} a_{11} \det[e_1, A_2, A_3, A_4] &= a_{11} \det[e_1, a_{12}e_1 + a_{22}e_2 + a_{32}e_3 + a_{42}e_4, A_3, A_4] \\ &= a_{11}a_{12} \det[e_1, e_1, A_3, A_4] + a_{11}a_{22} \det[e_1, e_2, A_3, A_4] \\ &\quad + a_{11}a_{32} \det[e_1, e_3, A_3, A_4] + a_{11}a_{42} \det[e_1, e_4, A_3, A_4]. \end{aligned}$$

Note that in the first summand, we have $\det[e_1, e_1, A_3, A_4]$, which is zero by the properties given in Definition 3.3.1. We continue like this and write each of the determinants involving an A_i as a linear combination of determinants involving e_i's. As soon as we see a determinant involving e_i's where there are two identical columns, for example $\det[e_1, e_1, A_3, A_4]$, we know that we get a zero. The only nonzero ones will be the determinants which contain all of the e_1, e_2, e_3, e_4.

To get such a determinant, we should be choosing a different row for each of the columns A_i. For example, above we got a summand $a_{11}a_{32}\det[e_1, e_3, A_3, A_4]$. Here, if we go back and reread the proof, we see that the e_1 appears in the first column because of the term a_{11}: the first row entry in the first column. And e_3 appears in the second column because it comes from the entry a_{32} in the third row of the second column.

But choosing a different row for each of the columns just means that we should have a bijective choice function C from the set of columns to the set of rows. But this is just a bijective function from the set of four elements to the set of four elements. This is, by definition, a permutation! So, for each permutation $\sigma \in S_4$, we will receive a nonzero determinant given by the columns e_1, e_2, e_3, e_4 possibly in a different order, that is, we will get $\det(\sigma I) = \operatorname{sgn}(\sigma)$ (see the proof of Proposition 3.4.10 for this notation).

This finishes the proof. □

Exercise 3.4.14. Go over the proof one more time and write it for general n.

3.5 The Product Formula

In this section, we are going to prove Theorem 3.1.1.

Chapter 4

Eigenthings

In this chapter, we are going to talk about eigenthings of a function: eigenvectors, eigenvalues, eigenspaces etc. The following exercise is the motivating factor.

Exercise 4.0.1. Compute A^5 and B^5 where

$$A = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 2 & 2 & 2 & 1 \\ 3 & 1 & -2 & 0 \\ 2 & 2 & 1 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

Do not skip this exercise. It will help you understand the reason why we care about eigenthings.

The idea is that it is easy to multiply diagonal matrices. It will probably take you at least 3 minutes to compute A^5 in the above exercise whereas if you know what you are doing, it takes less than a few seconds to compute B^5.

4.1 Eigenvalues and eigenvectors of a linear transformation

Now, assume that we have a vectorspace V with basis $\mathsf{B} = \{v_1, \dots, v_n\}$ and a linear transformation $f\colon V \to V$. As above, we would like to compute f^5. Remember that f^5 means composing 5 of this function f: That is, $f^5 = f \circ f \circ f \circ f \circ f$. One way to compute $f^5(v)$ for some $v \in V$ is to compute the representations with respect to the basis B and then using those representations to carry out the computation. More precisely, we know

$$[f^5(v)]_{\mathsf{B}} = [f^5]_{\mathsf{B},\mathsf{B}}[v]_{\mathsf{B}} = [f]^5_{\mathsf{B},\mathsf{B}}[v]_{\mathsf{B}}.$$

From the motivating exercise, we know that the matrix $[f]^5_{\mathsf{B},\mathsf{B}}$ is easy to compute if $[f]_{\mathsf{B},\mathsf{B}}$ is a diagonal matrix. Note that, if we change the basis B, the matrix representation usually changes, too. So, we have some choice here. So, the question is as follows.

Question 4.1.1. *Let V be a finite dimensional vectorspace over a field K. Given $f: V \to V$ linear. Can we choose a basis B of V so that the matrix representation $[f]_{\mathsf{B},\mathsf{B}}$ is diagonal?*

To answer this question, we rewrite it in several different ways. The following sentences are all equivalent.

1. There exists a basis $\mathsf{B} = \{v_1, \dots, v_n\}$ of V such that $[f]_{\mathsf{B},\mathsf{B}}$ is a diagonal matrix.
2. There exists a basis $\mathsf{B} = \{v_1, \dots, v_n\}$ of V such that
$$[f]_{\mathsf{B},\mathsf{B}} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$$
for some scalars $d_1, \dots, d_n \in K$.
3. There exists a basis $\mathsf{B} = \{v_1, \dots, v_n\}$ of V such that
$$\begin{aligned} f(v_1) &= d_1 v_1 + 0 \cdot v_2 + \cdots + 0 \cdot v_n \\ f(v_2) &= 0 \cdot v_1 + d_2 v_2 + \cdots + 0 \cdot v_n \\ \vdots &= \vdots \quad \vdots \quad \cdots \quad \vdots \\ f(v_n) &= 0 \cdot v_1 + 0 \cdot v_2 + \cdots + d_n v_n \end{aligned}$$
for some scalars $d_1, \dots, d_n \in K$.
4. There exists a basis $\mathsf{B} = \{v_1, \dots, v_n\}$ of V such that
$$f(v_1) = d_1 v_1 \ , \ f(v_2) = d_2 v_2 \ , \ \cdots \ , \ f(v_n) = d_n v_n.$$
for some scalars $d_1, \dots, d_n \in K$.

So, Question 4.1.1 becomes

Question 4.1.2. *Let V be a finite dimensional vectorspace over a field K and $f: V \to V$ a linear transformation. Can we find a basis $\mathsf{B} = \{v_1, \dots, v_n\}$ of V such that*
$$f(v_1) = d_1 v_1 \ , \ f(v_2) = d_2 v_2 \ , \ \cdots \ , \ f(v_n) = d_n v_n.$$
for some scalars $d_1, \dots, d_n \in K$?

We are now going to give this property a name.

Definition 4.1.3. Let V be a vectorspace over a field K and $f: V \to V$ linear. A scalar $c \in K$ is called an *eigenvalue* of f if there exists a nonzero vector $v \in V$ such that
$$f(v) = cv.$$
In this case, the nonzero vector v is called an *eigenvector* of f with eigenvalue c.

With this new words that we learned, we can rewrite Question 4.1.2.

Question 4.1.4. *Let V be a finite dimensional vectorspace over a field K and $f\colon V \to V$ a linear transformation. Can we find a basis of V consisting of eigenvectors of f?*

Let us see some examples of eigenvalues and eigenvectors.

Example 4.1.5. Consider the vectorspace $\mathrm{Diff}^\infty([0,1])$ of functions from $[0,1]$ to $\mathbb{R}$ which are infinitely many times differentiable. Let D be the function

$$\begin{aligned} D\colon \mathrm{Diff}^\infty([0,1]) &\to \mathrm{Diff}^\infty([0,1]) \\ f &\mapsto f' \end{aligned}$$

be the derivative operator. Then, we have

$$D(e^x) = e^x = 1 \cdot e^x$$

and so e^x is an eigenvector of D with eigenvalue 1. On the other hand, we have

$$D^2(\sin x) = D(\cos x) = -\sin x$$

which tells us that $\sin x$ is an eigenvector of D^2 with eigenvalue -1.

There are cases where Question 4.1.4 has a positive answer.

Example 4.1.6. Consider the linear transformation $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by the rule

$$f\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4x - y \\ 2x + y \end{bmatrix}.$$

Then, we have

$$f\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

and

$$f\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Then, the vector

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is an eigenvector of f with eigenvalue 3 and the vector

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

is an eigenvector of f with eigenvalue 2. These two vectors are linearly independent and form a basis of $\mathbb{R}^2$ and thus for the linear transformation in this example, Question 4.1.4 has an affirmative answer.

We are now going to see that the answer is not always affirmative.

Example 4.1.7. Consider the linear transformation

$$f\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$

from $\mathbb{R}^2$ to $\mathbb{R}^2$. Assume that c is an eigenvalue of this function f and that

$$\begin{bmatrix} a \\ b \end{bmatrix}$$

is an eigenvector f with eigenvalue c. Then, we have

$$f\begin{bmatrix} a \\ b \end{bmatrix} = c\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -b \\ a \end{bmatrix}$$

which gives us the equations

$$ca = -b \text{ and } cb = a.$$

From this we can conclude

$$c^2a = c(ca) = c(-b) = -bc = -a$$

and this gives us $(c^2 + 1)a = 0$. Similarly, we get

$$c^2b = c(cb) = ca = -b$$

which tells us that $(c^2 + 1)b = 0$. Since we are over real numbers, we know that there does not exist a $c \in \mathbb{R}$ such that $c^2 + 1 = 0$. This forces us to have $a = 0$ and $b = 0$. But we know from our definition that an eigenvector must be nonzero. So, this function f does not have an eigenvector or an eigenvalue which means that there is no basis of $\mathbb{R}^2$ consisting of eigenvectors of f, providing a negative answer to Question 4.1.4.

In this case, the negative answer came from the fact that the equation $c^2 + 1 = 0$ does not have any solutions, so we have no eigenvalues. But it is possible that there are eigenvalues of a linear transformation but not *enough* linearly independent eigenvectors to form a basis, as we see next.

Example 4.1.8. Consider the linear transformation

$$f\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ x \end{bmatrix}$$

from $\mathbb{R}^2$ to $\mathbb{R}^2$. Assume that c is an eigenvalue of this function f and that

$$\begin{bmatrix} a \\ b \end{bmatrix}$$

is an eigenvector f with eigenvalue c. Then, we have

$$f\begin{bmatrix}a\\b\end{bmatrix} = c\begin{bmatrix}a\\b\end{bmatrix} = \begin{bmatrix}0\\a\end{bmatrix}$$

which gives us the equations

$$ca = 0 \text{ and } cb = a.$$

Assume that $c \neq 0$. Then, the first equation tells us that $a = 0$. Then, the second equation tells us that $cb = 0$. Since $c \neq 0$, this tells us that $b = 0$. But then again, an eigenvector must be a nonzero vector and we can not have both $a = 0$ and $b = 0$. Therefore, we can NOT have $c = 0$. We must have $c = 0$.

In this case, the equation $ca = 0$ does not give us any new information and $cb = a$ tells us that we must have $a = 0$. We do not have any restrictions on b. So,

$$\begin{bmatrix}a\\b\end{bmatrix}$$

is an eigenvector of f if and only if $a = 0$. So, any eigenvector of f will look like

$$\begin{bmatrix}0\\b\end{bmatrix}$$

but any two such vectors are linearly dependent and therefore they can not form a basis. This means that there is no basis of $\mathbb{R}^2$ consisting of eigenvectors of f, providing a negative answer to Question 4.1.4.

We give a special name to the linear transformations for which Question 4.1.4 have a positive answer.

Definition 4.1.9. Let V be a finite dimensional vectorspace over a field K. We say that f is *diagonalisable* if there exists a basis of V consisting of eigenvectors of f.

Exercise 4.1.10. By comparing Question 4.1.1, Question 4.1.2 and Question 4.1.4, explain where the name "diagonalisable" comes from.

4.2 Eigenvalues and eigenvectors of a square matrix

Now, assume that K is a field and A is an $n \times n$ matrix with entries from the field K. Then, the matrix A defines a linear transformation

$$\begin{aligned} K^n &\to K^n \\ v &\mapsto Av. \end{aligned}$$

This point of view gives us the definition of eigenvalue/eigenvector of a matrix.

Definition 4.2.1. Let K be a field and A an $n \times n$ matrix with entries from K. A scalar $c \in K$ is called an *eigenvalue* of A if there exists a nonzero vector $v \in K^n$ such that

$$Av = cv.$$

In this case, the nonzero vector v is called an *eigenvector* of A with eigenvalue c. We say that A is diagonalisable if there exists a basis of K^n consisting of eigenvectors of A.

Exercise 4.2.2. Show that the vectors

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

are eigenvectors of the matrix

$$A = \begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}$$

and conclude that A is diagonalisable.

Exercise 4.2.3. Show that the following matrices are not diagonalisable:

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

4.3 The characteristic polynomial

In this section, we are going to talk about a method to find eigenvalues of a linear transformation or a matrix. There is one fact that we need to remember about invertible linear transformations:

1. A linear transformation $f \colon V \to V$ is invertible if and only if there does not exist a nonzero vector in the kernel of f.

2. A linear transformation $f \colon V \to V$ on a finite dimensional vectorspace is invertible if and only if the determinant of f is nonzero.

Ok, maybe that was two facts. But we need them. Now, assume that V is a finite dimensional vectorspace and $f \colon V \to V$ is a linear transformation. The following sentences are equivalent:

1. c is an eigenvalue of f.

2. There exists a nonzero vector v such that $f(v) = cv$.

3. There exists a nonzero vector v such that $cv - f(v) = 0$.

4. There exists a nonzero vector v such that $c \cdot \mathrm{id}(v) - f(v) = 0$.

5. There exists a nonzero vector v such that $(c \cdot \mathrm{id} - f)(v) = 0$.

6. The linear transformation $c \cdot \mathrm{id} - f$ is not invertible.

7. $\det(c \cdot \mathrm{id} - f) = 0$.

So, we proved the following.

Theorem 4.3.1. *Let V be a finite dimensional vectorspace over a field K and $f\colon V \to V$ linear. Then, $c \in K$ is an eigenvalue of f if and only if we have* $\det(c \cdot \mathrm{id} - f) = 0$.

Example 4.3.2. Previously, we considered the function $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4x - y \\ 2x + y \end{bmatrix}.$$

We showed that 2 and 3 are eigenvalues of f. We will find these eigenvalues now using the determinant method. The linear function $c \cdot \mathrm{id} - f$ is given by the rule

$$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} cx - (4x - y) \\ cy - (2x + y) \end{bmatrix} = \begin{bmatrix} (c-4)x + y \\ -2x + (c-1)y \end{bmatrix}.$$

The determinant of this function is the determinant of any matrix representation. With respect to the standard basis, the matrix represenation is

$$\begin{bmatrix} c - 4 & 1 \\ -2 & c - 1 \end{bmatrix}$$

and this has determinant $(c-1)(c-4) + 2 = c^2 - 5c + 6$. So, Theorem 4.3.1 tells us that c is an eigenvalue if and only if c is a root of the polynomial $p_f(t) = t^2 - 5t + 6$. Factorising, we get

$$p_f(t) = (t-2)(t-3)$$

and $(t-2)(t-3) = 0$ if and only if $t = 2$ or $t = 3$.

Example 4.3.3. We have seen that the matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

is not diagonalisable because it does not have any eigenvalues. Let us see this using the determinant method. We need to look at the polynomial

$$\begin{aligned} \det(t \cdot \mathrm{id} - A) &= \det\left(t \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \right) \\ &= \det \begin{bmatrix} t & 1 \\ -1 & t \end{bmatrix} \\ &= t^2 + 1 \end{aligned}$$

The polynomial equation $t^2 + 1 = 0$ does not have any real roots and therefore A is not diagonalisable over $\mathbb{R}$.

Example 4.3.4. We have seen that the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

has only one eigenvalue: namely, 0. Let us see this using the determinant method. We have

$$t \cdot \mathrm{id} - A = \begin{bmatrix} t & -1 \\ 0 & t \end{bmatrix}$$

and it has determinant t^2. We have $t^2 = 0$ if and only if $t = 0$ and this is the only eigenvalue.

Definition 4.3.5. Let $f\colon V \to V$ a linear transformation on a finite dimensional vectorspace. Then, the polynomial $\det(t \cdot \mathrm{id} - f)$ is called the *characteristic polynomial* of f.

Definition 4.3.6. Let A be an $n \times n$ matrix. Then, the polynomial $\det(t \cdot \mathrm{id} - A)$ is called the *characteristic polynomial* of A.

With this new word, we can write the main theorem of this section as follows.

Theorem 4.3.7. *Let $f\colon V \to V$ a linear transformation on a finite dimensional vectorspace and $p_f(t)$ be its characteristic polynomial. Then, c is an eigenvalue of f if and only if c is a root of the polynomial $p_f(t)$.*

Exercise 4.3.8. Prove this theorem. We have already discussed this, write the proof with your own words.

4.4 Eigenspaces

In this section, we are going to talk about diagonalisability in a more theoretic way. We start with a basic observation. Let $f\colon V \to V$ be a linear transformation on a finite dimensional vector space. The following sentences are equivalent.

1. v is an eigenvector of f with eigenvalue c.
2. v is a nonzero vector such that $f(v) = cv$.
3. v is a nonzero vector such that $cv - f(v) = 0$.
4. v is a nonzero vector such that $(c \cdot \mathrm{id} - f)(v) = 0$.
5. v is a nonzero vector in $\ker(c \cdot \mathrm{id} - f)$.

Definition 4.4.1. Let V be a finite dimensional vectorspace and $f\colon V \to V$ is a linear transformation with an eigenvalue c. We call the subspace $\ker(c \cdot \mathrm{id} - f)$ the *eigenspace* of f for the eigenvalue c.

Exercise 4.4.2. Let V be a finite dimensional vectorspace and $f\colon V \to V$ is a linear transformation with an eigenvalue c. Let E_c be the eigenspace of f for the eigenvalue of c. Show that $\dim(E_c) \geq 1$.

Let us now recall some facts about direct sum decompositions.

1. Let V be a vectorspace and U, W be subspaces. We say that V is a direct sum of U and W and write $V = U \oplus W$ if the following two conditions are satisfied:
   (a) Any $v \in V$ can be written as $v = u + w$ with $u \in U$ and $w \in W$, and
   (b) $U \cap W = \{0\}$.
2. Let V be a vectorspace and $U_1, \dots, U_\ell$ be subspaces. We write
$$V = U_1 \oplus \cdots \oplus U_\ell$$
   if the following two conditions are satisfied:
   (a) Any $v \in V$ can be written as $v = u_1 + \cdots + u_\ell$ with $u_i \in U_i$.
   (b) For $i \neq j$, we have $U_i \cap U_j = \{0\}$.
3. Assume $V = U_1 \oplus \cdots \oplus U_\ell$. Let B_i be a basis for each subspace U_i for $i = 1, \dots, \ell$. Then, the union
$$\mathsf{B} = \mathsf{B}_1 \cup \cdots \cup \mathsf{B}_\ell$$
   is a basis for the whole space.

With this, we are now ready to rewrite diagonalisation in a new way.

Theorem 4.4.3. *Let V be a finite dimensional vectorspace and $f: V \to V$ is a linear transformation with eigenvalues $c_1, \dots, c_\ell$. Then, f is diagonalisable if and only if we have a direct sum decomposition*
$$V = E_{c_1} \oplus \cdots \oplus E_{c_\ell}$$
where E_{c_i} denotes the eigenspace for the eigenvalue c_i.

Exercise 4.4.4. Prove this theorem. The proof should follow after writing down the definitions and using the facts above about direct sum decompositions. The only thing you need to prove yourself is the following: if c, d are two distinct eigenvalues, then the eigenspace E_c and E_d satisfy $E_c \cap E_d = \{0\}$.

We are now going to explore this more. We need the following fact: Let $U_1, \cdots, U_\ell$ be subspaces of a finite dimensional vectorspace V with $U_i \cap U_j = \{0\}$ for $i \neq j$. Then, $\dim(U_1 + \cdots + U_\ell) = \dim U_1 + \cdots + \dim U_\ell$. In particular, we have
$$\dim U_1 + \cdots + \dim U_\ell = n \text{ if } V = U_1 \oplus \cdots \oplus U_\ell.$$

We already know that distinct eigenspaces intersect only at 0. Hence, we conclude the following theorem.

Theorem 4.4.5. *Let V be a finite dimensional vectorspace and $f: V \to V$ is a linear transformation with eigenvalues $c_1, \dots, c_\ell$. Then, f is diagonalisable if and only if we have*
$$\dim E_{c_1} + \cdots + \dim E_{c_\ell} = \dim V$$
where E_{c_i} denotes the eigenspace for the eigenvalue c_i.

We give these dimensions a special name.

Definition 4.4.6. Let V be a finite dimensional vectorspace and $f: V \to V$ is a linear transformation. Assume that c is an eigenvalue of f.

1. We say that c has *algebraic multiplicity* m if we have $p_f(t) = (t-c)^m q(t)$ and $t-c$ does not divide $q(t)$, where p_f is the characteristic polynomial of f.

2. We call $\dim(E_c)$ the *geometric multiplicity* of c.

The following theorem follows from considering quotient spaces easily but we are skipping the proof.

Theorem 4.4.7. *Let V be a finite dimensional vectorspace and $f: V \to V$ is a linear transformation. Assume that c is an eigenvalue of f. Then, the geometric multiplicity of c is bounded above by the algebraic multiplicity of c.*

And now, we are ready to state the main theorem of this section.

Theorem 4.4.8. *Let V be a finite dimensional vectorspace and $f: V \to V$ is a linear transformation. Assume that $c_1, \ldots, c_\ell$ are a complete set of distinct eigenvalues of f with algebraic multiplicities $n_1, \ldots, n_\ell$ and geometric multiplicities $m_1, \ldots, m_\ell$ respectively. Then, f is diagonalisable if and only if the following two conditions hold:*

1. $n_1 + \cdots + n_\ell = \dim V$,

2. $n_i = m_i$ *for each* $i = 1, \ldots, \ell$.

Remark 4.4.9. The first condition is telling us that the characteristic polynomial of f *splits.* This is automatic in an *algebraically closed* field where every polynomial has at least one root. For instance, over complex numbers, we do not neet to worry about this condition. But the function

$$f \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix}$$

on $\mathbb{R}^2$ has characteristic polynomial t^2+1 and it does not have any roots, so it fails to satisfy the first condition of the theorem.

Remark 4.4.10. The first condition is satisfied for the function

$$f \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ x \end{bmatrix}$$

on $\mathbb{R}^2$. It has characteristic polynomial t^2. The eigenvalue 0 has algebraic multiplicity 2. But the geometric multiplicity is only 1 because we have

$$E_0 = \left\{ \begin{bmatrix} 0 \\ y \end{bmatrix} : y \in \mathbb{R} \right\}$$

as the eigenspace for 0 and this is a one dimensional subspace.

4.5 Diagonalisation

In this section, we are going to talk about *how to diagonalise* a matrix if we already know that it is diagonalisable. What do we mean by this?

Let A be an $n \times n$ matrix with coefficients from a field K. As discussed before, A defines a linear transformation

$$T_A \colon K^n \to K^n$$

given by the rule

$$T_A(v) = Av.$$

The following exercise follows from just looking at the definitions and it is only saying a very obvious fact in some weird way. If you are not comfortable with it, prove it and see it on an exercise.

Exercise 4.5.1. Let A and T_A be as above and let std denote the standard basis $\{e_1, \ldots, e_n\}$ of K^n. Then, we have

$$[T_A]_{\text{std},\text{std}} = A.$$

The following is also straightforward if you are comfortable with matrix representations.

Exercise 4.5.2. Let $v \in K^n$. Then, v is an eigenvector of A if and only if it is an eigenvector of T_A.

The following is once again straightforward if you are comfortable with change of basis matrices.

Exercise 4.5.3. Let $\mathsf{B} = \{v_1, \ldots, v_n\}$ be a basis of K^n. Then, the change of basis matrix $[\text{id}]_{\mathsf{B},\text{std}}$ is the matrix whose columns are $v_1, \ldots, v_n$.

Now, using all of these exercises and remembering one important fact about (will you be able to see which important fact?) change of basis matrices from Chapter 1, we arrive at the following theorem.

Theorem 4.5.4. *Let A be an $n \times n$ diagonalisable matrix with coefficients from a field K. Let S be the matrix whose columns are linearly independent eigenvectors of f. Then, we have*

$$D = S^{-1}AS$$

where D is the diagonal matrix with eigenvalues on the diagonal.

We finish this chapter with an application on Fibonacci numbers.

Chapter 5

Normal maps

We started this semester with matrix representations and continued with inner product spaces and the eigenvalue theory. We have learned the following.

1. If I have a finite dimensional vectorspace and an inner product on it, then the *desirable* bases are the orthonormal ones.

2. If I have a finite dimensional vectorspace and a linear function on it, then the *desirable* bases are the ones which consist of eigenvectors of the function.

Now, the natural thing to do is to ask what happens if we combine the two stories.

Question 5.0.1. *If I am given a vectorspace, an inner product on it and a linear transformation, can I find a basis which is both orthonormal and also consisting of eigenvectors of the linear transformation?*

In this section, we are interested in this question.

Example 5.0.2. We start with an example where the question has a positive answer. Consider the identity transformation $\mathrm{id}\colon \mathbb{R}^2 \to \mathbb{R}^2$. Let $\{e_1, e_2\}$ be the standard basis. Then, this is an orthonormal basis and since $\mathrm{id}(e_1) = e_1 = 1 \cdot e_1$ and $\mathrm{id}(e_2) = e_2 = 1 \cdot e_2$, both e_1 and e_2 are eigenvectors with eigenvalue 1. So, $\{e_1, e_2\}$ is an orthonormal basis of $\mathbb{R}^2$ consisting of eigenvectors of id.

On the other hand, one can find examples where we have a negative answer.

Example 5.0.3. Consider the function

$$f \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2y \\ 0.5x \end{bmatrix}$$

on $\mathbb{R}^2$ with the standard inner product. Then, we have

$$f \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and } f \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

So, there are two eigenvalues 1 and -1 but the eigenvectors

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

are not orthogonal to each other. So, we do not have an orthonormal basis consisting of eigenvectors of f.

We will prove the following theorem in the next section.

Theorem 5.0.4. *Let V be a finite dimensional unitary space and $f: V \to V$ a linear transformation. Then, there exists an orthonormal basis of V consisting of eigenvectors of f if and only if f is normal.*

Of course, to do this, we need to learn what normal means.

5.1 Normal maps

One can write Theorem 5.0.4 as the definition of a normal map on a unitary space, of course. But our goal is to prove it as a theorem by starting from a different definition.

Definition 5.1.1. Let V be a Euklidean or unitary space and $f: V \to V$ a linear transformation. We call f *normal* if the equation $f \circ f^* = f^* \circ f$ holds. A square matrix A is called *normal* if A commutes with A^*.

Exercise 5.1.2. Consider the function

$$f \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2y \\ 0.5x \end{bmatrix}$$

on $\mathbb{R}^2$ with the standard inner product. Compute f^*. Then, show that $f \circ f^* \neq f^* \circ f$.

Our goal is to prove Theorem 5.0.4. We will do this by exploring different properties of normal maps.

Theorem 5.1.3. *A linear transformation $f: V \to V$ is normal if and only if for every $x, y \in V$ one has*

$$\langle f^*(x), f^*(y) \rangle = \langle f(x), f(y) \rangle.$$

Exercise 5.1.4. Prove this theorem. All you need is the definition of the adjoint map and the fact that $(f^*)^* = f$.

Exercise 5.1.5. Show that if $f: V \to V$ is a normal map, then for any $v \in V$ we have

$$\|f(v)\| = \|f^*(v)\|.$$

Lemma 5.1.6. *Let $f: V \to V$ be a normal map. Then, there is an equality*

$$\ker(f) = \ker(f^*).$$

Proof. Let $v \in \ker(f)$. Then, $f(v) = 0$ and therefore we have $\|f(v)\| = 0$. Then, we have $\|f^*(v)\| = 0$ and this implies $f^*(v) = 0$. This shows the inclusion $\ker(f) \subseteq \ker(f^*)$. The other inclusion follows from this by using the fact that $(f^*)^* = f$. ☐

Lemma 5.1.7. *Let $f\colon V \to V$ be a normal map. Then, $f + \mathrm{id}$ is also normal.*

Proof. We know that $(f + \mathrm{id})^* = f^* + \mathrm{id}^* = f^* + \mathrm{id}$. Then, we have

$$(f + \mathrm{id}) \circ (f + \mathrm{id})^* = (f + \mathrm{id}) \circ (f^* + \mathrm{id}) = ff^* + f + f^* + \mathrm{id} \text{ and}$$
$$(f + \mathrm{id})^* \circ (f + \mathrm{id}) = (f^* + \mathrm{id}) \circ (f + \mathrm{id}) = f^*f + f + f^* + \mathrm{id}.$$

Since $ff^* = f^*f$, these two are equal. This finishes the proof. ☐

Exercise 5.1.8. Let $f\colon V \to V$ be a normal map. Then, $f + c \cdot \mathrm{id}$ is also normal for any scalar c.

Lemma 5.1.9. *Let $f\colon V \to V$ be a normal map. Assume that c is an eigenvalue of f with eigenvector v. Then, v is also an eigenvector of f^* with eigenvalue $\bar{c}$.*

Proof. We know that a nonzero vector v is an eigenvector with eigenvalue c if and only if $v \in \ker(c \cdot \mathrm{id} - f)$. By the previous exercise, we know that $c \cdot \mathrm{id} - f$ is normal. Then, by Lemma 5.1.6, we get that $v \in \ker(c \cdot \mathrm{id} - f)^* = \ker(\bar{c} \cdot \mathrm{id} - f^*)$. This finishes the proof. ☐

The next proposition is the key to achieve our theorem.‘

Proposition 5.1.10. *Let $f\colon V \to V$ be a normal linear transformation and c_1, c_2 two distinct eigenvalues with eigenspaces E_1, E_2 respectively. Then, we have $E_1 \perp E_2$.*

Proof. We need to show that $\langle v_1, v_2 \rangle = 0$ for any $v_1 \in E_1$ and $v_2 \in E_2$. So, let $v_1 \in E_1$ and $v_2 \in E_2$. Then, we have

$$c_2\langle v_1, v_2 \rangle = \langle v_1, c_2v_2 \rangle = \langle v_1, f(v_2) \rangle = \langle f^*(v_1), v_2 \rangle = \langle \bar{c}_1v_1, v_2 \rangle = c_1\langle v_1, v_2 \rangle.$$

This gives us

$$c_2\langle v_1, v_2 \rangle = c_1\langle v_1, v_2 \rangle$$

and therefore

$$(c_1 - c_2)\langle v_1, v_2 \rangle = 0.$$

Since $c_1 \neq c_2$, we deduce that $\langle v_1, v_2 \rangle = 0$. ☐

We are almost ready to prove Theorem 5.0.4. We need one more lemma.

Lemma 5.1.11. *Let $f\colon V \to V$ be a normal linear transformation. Let $v \in V$ be an eigenvector of f with eigenvalue c and put*

$$U = \{x \in V \colon \langle x, v \rangle = 0\}.$$

Then, for any $u \in U$, one has $f(u) \in U$. In other words, we have $f|_U\colon U \to U$.

Proof. Let $u \in U$. We want to show that $f(u) \in U$. This means that we should show $\langle f(u), v\rangle = 0$. This follows from

$$\langle f(u), v\rangle = \langle u, f^*(v)\rangle = \langle u, \bar{c}v\rangle = \bar{c}\langle u, v\rangle = 0.$$

□

Theorem 5.1.12. *Let V be a finite dimensional unitary space and $f\colon V \to V$ a linear transformation. Then, there exists an orthonormal basis of V consisting of eigenvectors of f if and only if f is normal.*

Proof. Assume that f is normal. We will show that there exists an orthonormal basis consisting of eigenvectors of f. Since V is a unitary space, we know that it is a vectorspace over complex numbers. Therefore, we consider the characteristic polynomial with values in complex numbers. By the fundamental theorem of algebra, every polynomial with complex coefficients has a root in complex numbers. So, f has at least one eigenvalue and one eigenvector ω with eigenvalue c. We will proceed by induction. If V is one dimensional, then $\{\omega/\|\omega\|\}$ is an orthonormal basis for V. Now, assume that $n \geq 2$ and V is n-dimensional. Assume that the statement we want to show is true for $n-1$ dimensional spaces. Put

$$U = \{x \in V\colon \langle x, \omega\rangle = 0\}.$$

Since $U = \operatorname{span}(\omega)^\perp$, it is an $n-1$ dimensional space. By the previous lemma, we know that $f|U$ is a normal function from U to U. So, U has an orthonormal basis $\{v_1, \ldots, v_{n-1}\}$ consisting of eigenvectors of $f|U$ which are also eigenvectors of f. Then, $\{\omega/\|\omega\|, v_1, \ldots, v_n\}$ is an orthonormal basis consisting of eigenvectors of f. This is what we wanted to show.

Now, we will show the other direction. Assume that there exists an orthonormal basis $\mathsf{B} = \{v_1, \ldots, v_n\}$ of V consisting of eigenvectors of f. We want to show that f is normal. Let $f(v_i) = c_i v_i$. Define a new linear transformation $g\colon V \to V$ with $g(v_i) = \bar{c}v_i$. Then, we have

$$\langle f(v_i), v_j\rangle = \langle c_i v_i, v_j\rangle = \bar{c_i}\langle v_i, v_j\rangle = \begin{cases} \bar{c} & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and

$$\langle v_i, g(v_j)\rangle = \langle v_i, \bar{c_j} v_j\rangle = \bar{c_j}\langle v_i, v_j\rangle = \begin{cases} \bar{c_j} & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

This shows that $g = f^*$. Moreover, we have

$$\begin{aligned} gf(v_i) &= g(c_i v_i) = c_i g(v_i) = c_i \bar{c_i} v_i \\ gf(v_i) &= f(\bar{c_i} v_i) = \bar{c_i} f(v_i) = \bar{c_i} c_i v_i. \end{aligned}$$

This shows that $gf = fg$ and this shows f is normal. □

Remark 5.1.13. It is important to note that the theorem tells us that normal maps are diagonalisable but not every diagonalisable transformation is normal.

The next theorem tells us what happens for Euklidean spaces.

Theorem 5.1.14. *Let V be an n-dimensional Euklidean vectorspace and $f: V \to V$ be a linear transformation. There exists an orthonormal basis of V consisting of eigenvectors of f if and only if f is normal and f has real eigenvalues.*

Exercise 5.1.15. The proof is exactly the same as the previous proof. Write it in your own words.

5.2 Self adjoint maps

The condition for being a normal map is that a function should commute with its adjoint. Given a function $f: V \to V$ on a unitary or a Euklidean space, not every other function g commutes with f. We can not always have $f: g = g \circ f$. In fact, one has $f \circ g = g \circ f$ for every g if and only if f is a multiple of the identity map. But we know that there are some functions which commute with a given function f. For instance, f commutes with itself. That is, $f \circ f = f \circ f$! So, if it was the case that $f^* = f$, then f would be normal! In this section, we are interested in this case.

Definition 5.2.1. Let V be a finite dimensional Euklidean or unitary space and $f: V \to V$ is a linear transformation. We call f *self-adjoint* if $f^* = f$.

Remark 5.2.2. We have seen Hermitian matrices before. If f is self-adjoint and B is an orthonormal basis, then $[f]_{\mathsf{B},\mathsf{B}}$ is a Hermitian matrix. We have also remarked this before.

Exercise 5.2.3. Show that a self-adjoint map is normal. We discussed this above. Use that idea and write a rigorous proof.

Theorem 5.2.4. *Let V be an n-dimensional Euklidean or unitary space. If $f: V \to V$ is a self-adjoint function, then f has n real eigenvalues.*

Proof. If V is Euklidean, we have already seen that it is diagonalisable. So, assume that V is unitary. Since f is normal, there exists an orthonormal basis B consisting of eigenvectors of f so that we have

$$[f]_{\mathsf{B},\mathsf{B}} = D := \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} = [f^*]_{\mathsf{B},\mathsf{B}}$$

where the last equality is because f is self-adjoint. On the other hand, we have

$$[f^*]_{\mathsf{B},\mathsf{B}} = [f]^*_{\mathsf{B},\mathsf{B}} = \begin{bmatrix} \overline{\lambda}_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \overline{\lambda}_n \end{bmatrix}.$$

Combining these we can deduce

$$\begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} = \begin{bmatrix} \overline{\lambda}_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \overline{\lambda}_n \end{bmatrix}.$$

$\square$

5.3 Unitary maps